

THERMABRICK · BUILD PLAN

ThermaBrick prototype build plan

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	First build plan, with pictures by component and step; design made constructable (TBK-DDR-003)

ThermaBrick prototype build plan

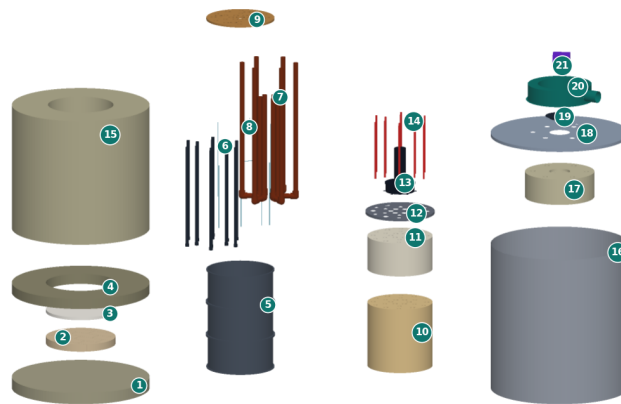
Plan, not yet built. How to build the first full-scale proof-of-concept ThermaBrick, component by component. Building and testing to it is TRL 4 work, which is on hold. Decisions still to be made are kept in the design decisions register ([docs/06-design-decisions.md](#)), not here.

■ 1. What you are building

ThermaBrick prototype: every component, pulled apart

Numbered in build order, in four columns: base, drum and pipes, what goes in and on the drum, the jacket. The template (9) is a temporary jig

- 1 Stone wool base boards (2 layers)
- 2 Firebrick disc
- 3 AES board disc
- 4 Batt ring round the disc
- 5 Drum, prepared
- 6 Heater wells (12)
- 7 U-tubes (6)
- 8 Thermocouples and guide rods
- 9 Setting template (removed after the fill)
- 10 Sand, 210 kg
- 11 Headspace insulation
- 12 Drum lid, drilled, and ring
- 13 Collector and outlet
- 14 Cartridge heaters (12)
- 15 Side insulation
- 16 Jacket side
- 17 Top insulation
- 18 Jacket cap
- 19 Outlet packing and trim ring
- 20 Inlet plenum
- 21 Heater junction box



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Figure 1. Every component pulled apart and numbered in build order, in four columns: the base; the drum and the pipes that stand in it; what goes into and onto the drum; the jacket and what sits on top. The setting template (9) is a temporary jig.

ThermaBrick is a 55 US gal steel drum filled with 210 kg of dry sand, with twelve electric heaters in capped steel pipes (the wells) to charge it and six steel U-tubes to carry room air through it and take the heat out. The drum stands on an insulated base and is wrapped in 317 mm of fiber insulation inside a galvanized steel jacket, 1,209 mm across and 1,462 mm high. Figure 1 shows the 21 components in the order you make or fit them. Ten are made in a home workshop: the base layers, the drilled drum lid, a plywood setting template, the wells, the U-tubes, the collector and outlet, the jacket side and cap, a steel trim ring and the inlet plenum. Everything else is bought: sand, heaters, thermocouples, insulation, the air path parts and the controls. The work is cutting and threading black steel pipe, drilling sheet steel with hole saws, cutting firebrick and fiber insulation, and cutting, folding and riveting thin sheet metal. No welding, machining or pressure parts. The parts cost about USD 3,850 from the bill of materials. The finished unit weighs about 415 kg and must stand on a concrete slab.

Safety: The bed runs at up to 550 °C at the heater wells and stays hot for days after the power is off. The heaters run on 240 V: the branch circuit and the control enclosure are wired or checked by a licensed electrician. Pouring sand and cutting firebrick release respirable crystalline silica, and fiber insulation irritates skin, eyes and airways: work outdoors or with local exhaust, and wear a P100 or N95 respirator, gloves, long sleeves and eye protection. The first firing gives off smoke from coatings and the insulation binder. Section 6 lists every stop point.

■ **2. What changed to make it buildable**

The concept showed what ThermaBrick does and sized it; some of its parts could not be made, fixed or assembled as drawn. Each change below keeps what the unit does, and all of them are recorded in decision record TBK-DDR-003, open for Amish's review.

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Heater wells	Wells hanging 14 mm above the drum floor with nothing under them	Each well stands on a levelled 14 mm first layer of sand, at the same height (Figure 9)	Supports the wells without putting hot steel against the drum floor
U-tube bottoms	Floating 50 mm above the floor; the listed 2-1/2 in nipple set the legs 118 mm apart, not 155 mm	Two elbows and a 4 in nipple, resting on the floor (Figure 11)	The stock fittings now give the 155 mm leg spacing, and the tube has a firm seat
Lid holes	1 mm clear of each pipe; no way through for the thermocouples	36 mm holes for the wells, 48 mm for the U-tube legs, 6 mm for four sand thermocouples, a fiber rope collar round each pipe (Figures 5, 6)	The lid can be lowered over 24 pipes at once; the well-wall thermocouples pass beside their wells
Holding the pipes during the fill	The lid on spacers, which leaves no way to pour the sand	A plywood setting template on the drum rim, with five pour holes (Figure 7)	Holds every pipe upright until the sand locks it
Base	One solid layer; firebrick only under the drum, leaving a gap out to the jacket	A 610 mm firebrick disc and AES board disc, with a ring of stone wool batt round them (Figures 2, 3, 4)	A level, fully filled base; the firebrick still carries the drum
Inlet plenum	110 mm tall, 4 mm of wall round the 4 in collar, no fixing	140 mm tall, collar hole cut through, flanges riveted to the cap (Figures 23, 24)	Room to rivet the collar; a fixing for the plenum
Collector	No fixing to the lid	Six tabs folded out between the wells and riveted to the lid (Figures 13, 14)	A fixing that clears the wells
Outlet through the jacket	Galvanized cap touching the hot outlet	A 180 mm hole packed with fiber and covered by a black steel trim ring (Figures 21, 22)	Keeps zinc away from the hot pipe, as requirement R13 asks
Heater leads and junction box	36 in leads, too short for most wells; no position for the box	72 in leads; the box on the cap at 120°, 430 mm out (Figures 19, 20)	Every lead reaches the box
Thermocouples	No route out of the drum; 1,000 mm sheaths too short	Routes through the lid, guide rods for the sand sensors, 1,500 mm sheaths, one exit in the cap (Figures 6, 12)	Every sensor reaches the outside at its stated measuring point
Jacket cap	No fixing; drawn as one sheet wider than the flashing	Three lapped strips with a turned-down skirt, held by screws (Figures 16, 18)	The cap can be lifted off to change a heater

■ 3. Making the components

Make and check each component before the assembly step that needs it. Sizes are in millimetres. Angles are measured anticlockwise, seen from above, from a 0° mark on the drum rim (the side where the inlet plenum's collar will point). Workshop tolerance is 2 mm unless a step says otherwise; drawings do not carry tolerances before TRL 4.

3.1 Base

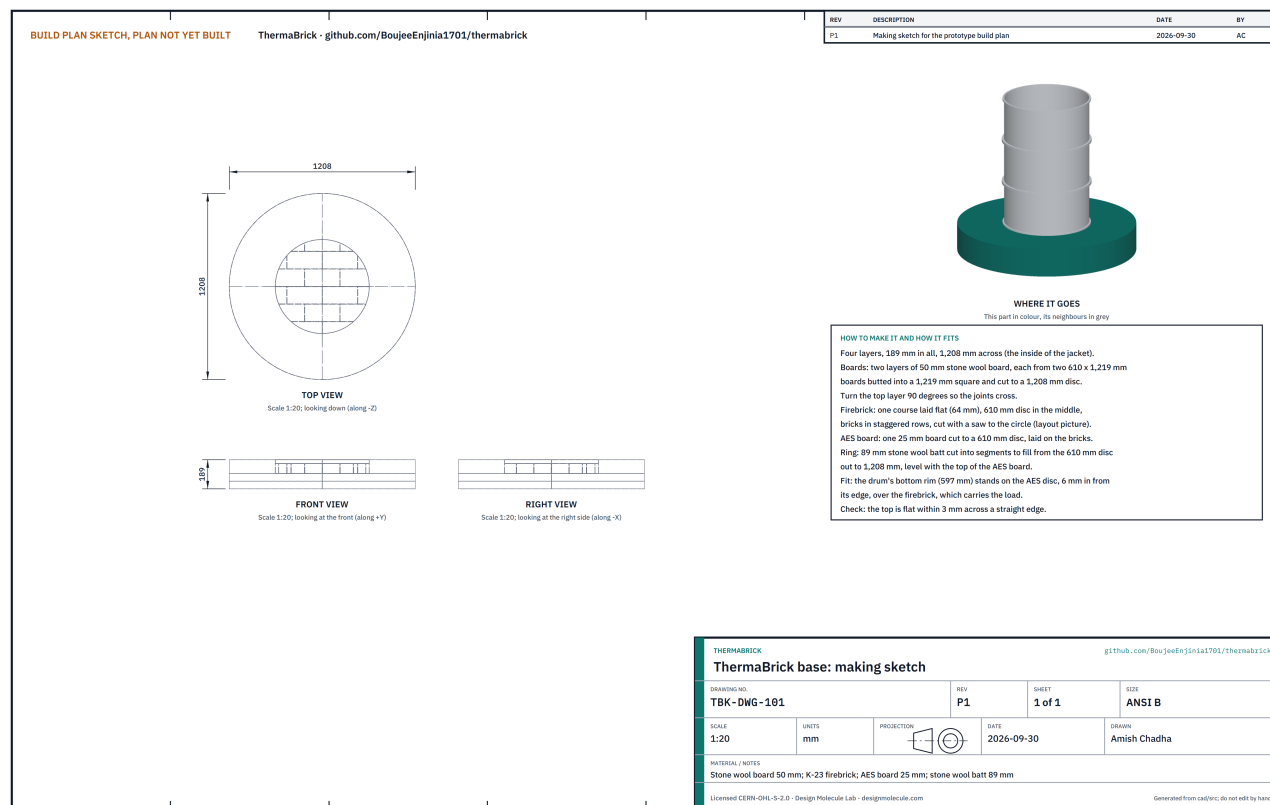
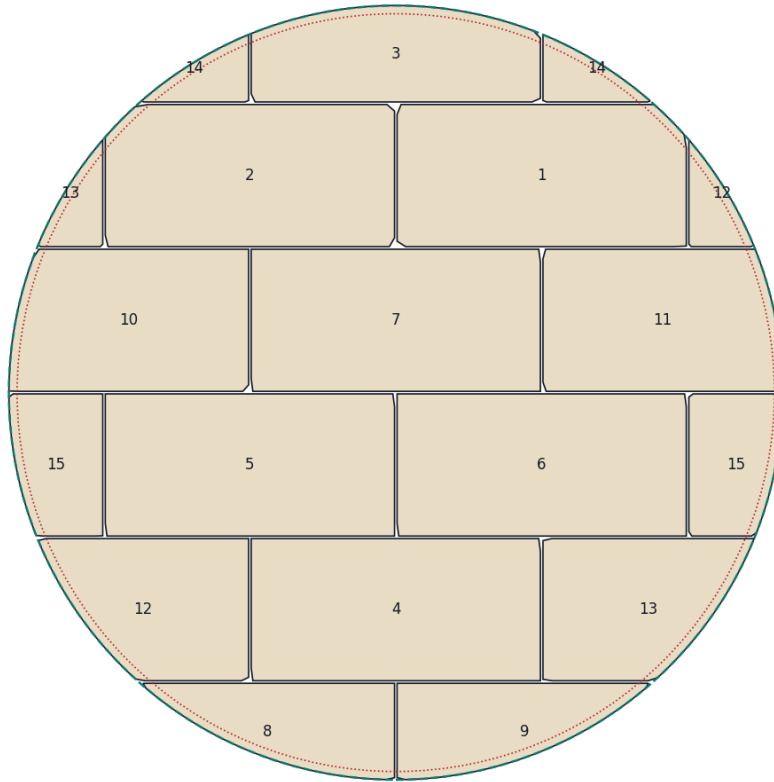


Figure 2. Base making sketch (TBK-DWG-101).

Firebrick disc: cutting layout

Seen from above. K-23 bricks 230 x 114 x 64 mm laid flat in staggered rows and cut to a 610 mm circle. Numbers are the bricks each piece is cut from.



19 pieces from 15 bricks;
buy 16 (one spare).

Lay the rows on the stone wool board, joints staggered by half a brick, 2 mm joints, no mortar.

Mark the 610 mm circle (dashed) with a trammel and saw to it with a coarse handsaw or a masonry blade, outdoors, with a P100 or N95 respirator.

Red dotted: the drum's 597 mm bottom rim, which stands on the AES disc above the bricks.

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Figure 3. Firebrick disc cutting layout: 19 pieces cut from 15 bricks.

What it is and what it is made from. Four layers, 189 mm thick in all, that carry the drum and keep its heat out of the floor. Two layers of 50 mm stone wool board (four boards of 610 x 1,219 mm); one course of K-23 insulating firebrick laid flat (16 bricks of 230 x 114 x 64 mm); one 25 mm AES fiber board (610 x 914 mm); and a ring of 89 mm stone wool batt.

How to make it.

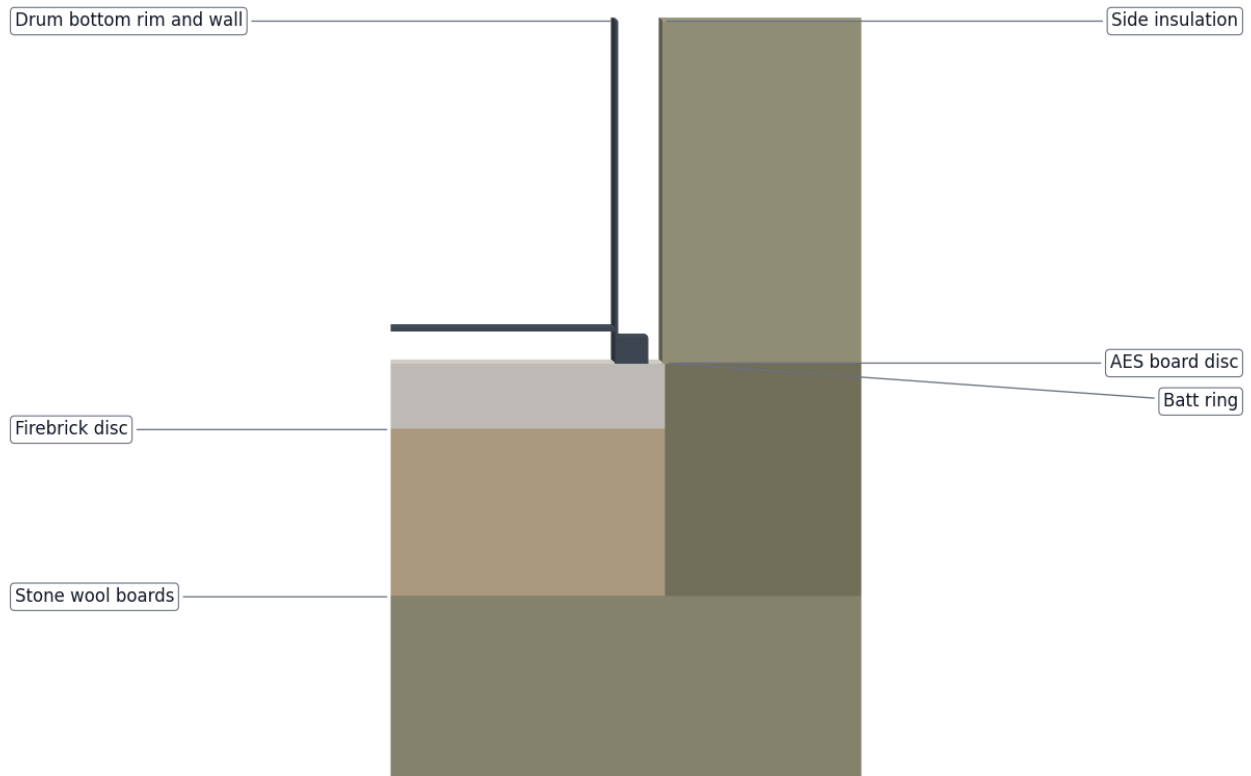
1. Butt two stone wool boards into a 1,219 mm square, draw a 1,208 mm circle with a trammel and cut it with a long serrated knife. Repeat for the second layer.
2. Lay the bricks dry in staggered rows as Figure 3 shows, with 2 mm joints. Mark a 610 mm circle and number the pieces.
3. Cut the bricks outdoors with a coarse handsaw or a masonry blade, wearing a respirator. Use the offcuts for the small edge pieces marked with the same number.
4. Cut a 610 mm disc from the AES board.

5. Cut the 89 mm batt into segments that fill the ring from 610 mm to 1,208 mm across.

How it fits the parts next to it.

Joint 1: the drum on the base (cut open at the edge)

Seen from the front. The drum's rim stands on the AES disc over the firebrick; the batt ring fills out to the jacket



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Figure 4. The drum's bottom rim stands on the AES disc, 6 mm in from its edge, over the firebrick; the batt ring fills out to the jacket, level with the AES disc.

The two board layers lie on the slab with their joints crossed. The firebrick disc sits in the middle of the top board, the AES disc on the bricks, and the batt ring round both, all level at 189 mm. The drum's 597 mm bottom rim stands on the AES disc; the firebrick under it carries the full 415 kg.

Check before moving on. A straight edge across the top shows no gap over 3 mm; the AES disc does not rock.

3.2 Drum and drum lid

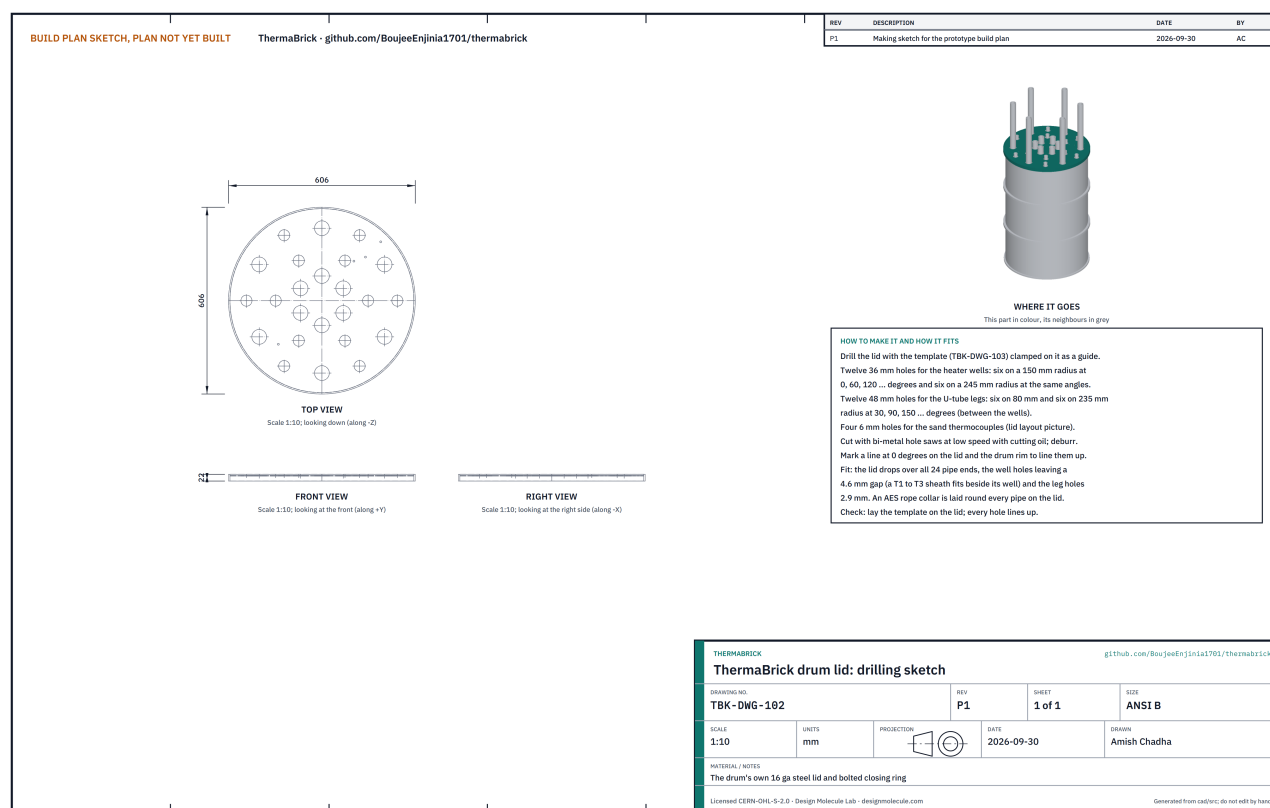
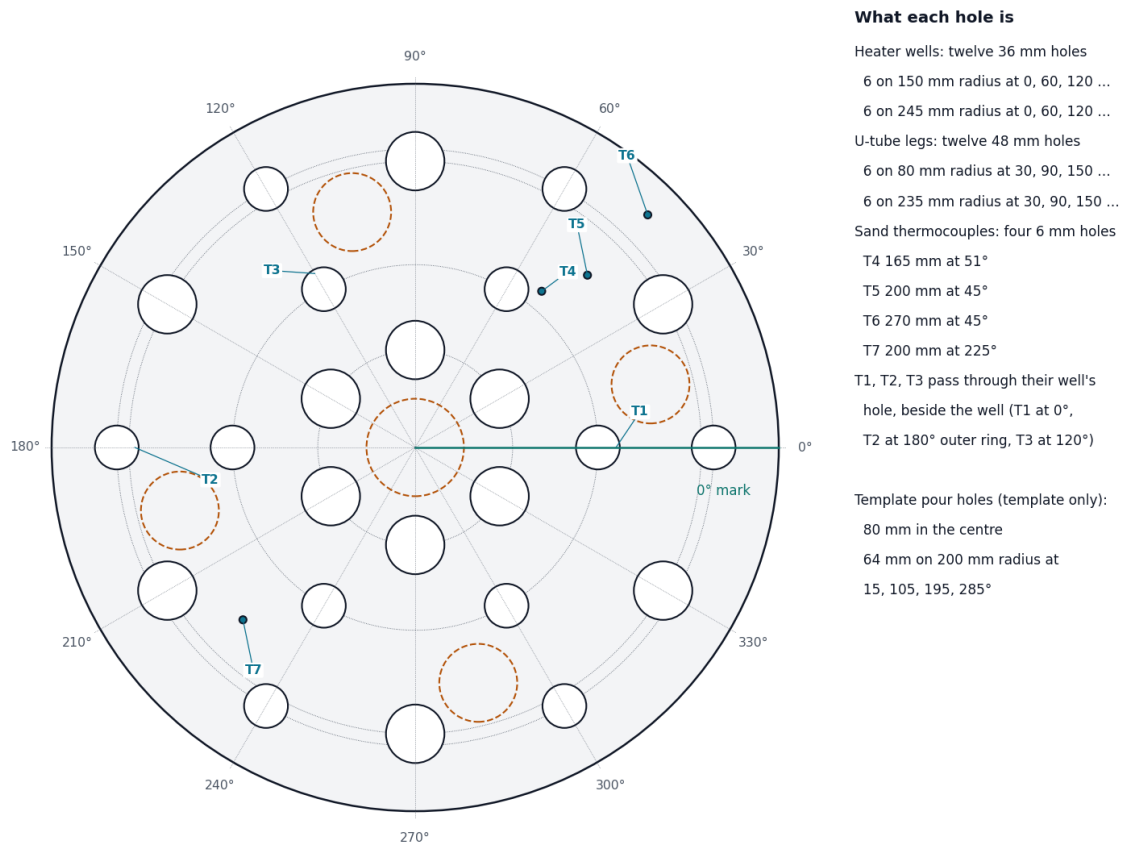


Figure 5. Drum lid drilling sketch (TBK-DWG-102).

Drum lid and setting template: hole layout

Seen from above. Radius from the centre and angle from the 0 degree mark, anticlockwise. Template only: the four 64 mm and the 80 mm pour holes (dashed).



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Figure 6. Lid and template hole layout, with the thermocouple positions T1 to T7.

What it is and what it is made from. A new, unlined, open-head 55 US gal steel drum (571.5 mm inside, 883 mm high, 18 ga body) with its 16 ga lid and bolted closing ring. The lid carries 28 holes.

How to make it.

1. Strip the outside paint with a flap disc. Burn the empty drum outdoors to about 300 °C (a wood fire round it) to remove residues; let it cool and brush it out.
2. Mark a 0° line on the drum rim and on the lid edge.
3. Make the setting template first (section 3.3), clamp it on the lid with the 0° marks together, and drill through it.
4. Twelve 36 mm holes for the wells: six on a 150 mm radius and six on 245 mm, both at 0, 60, 120, 180, 240 and 300°.
5. Twelve 48 mm holes for the U-tube legs: six on 80 mm and six on 235 mm, at 30, 90, 150, 210, 270 and 330°.
6. Four 6 mm holes for the sand thermocouples: T4 at 165 mm and 51°, T5 at 200 mm and 45°, T6 at 270 mm and 45°, T7 at 200 mm and 225°.

7. Use bi-metal hole saws at low speed with cutting oil; deburr and degrease.

How it fits the parts next to it. The lid drops over all 24 pipe ends at once: each well hole leaves 4.6 mm round the well, enough for the well-wall thermocouple beside it, and each leg hole 2.9 mm. A collar of 12 mm AES rope is laid round every pipe on top of the lid, and the closing ring is bolted as the drum maker supplies it. The bed breathes through the collars; nothing is sealed or pressurized.

Check before moving on. The template laid on the lid lines up with every hole.

3.3 Setting template

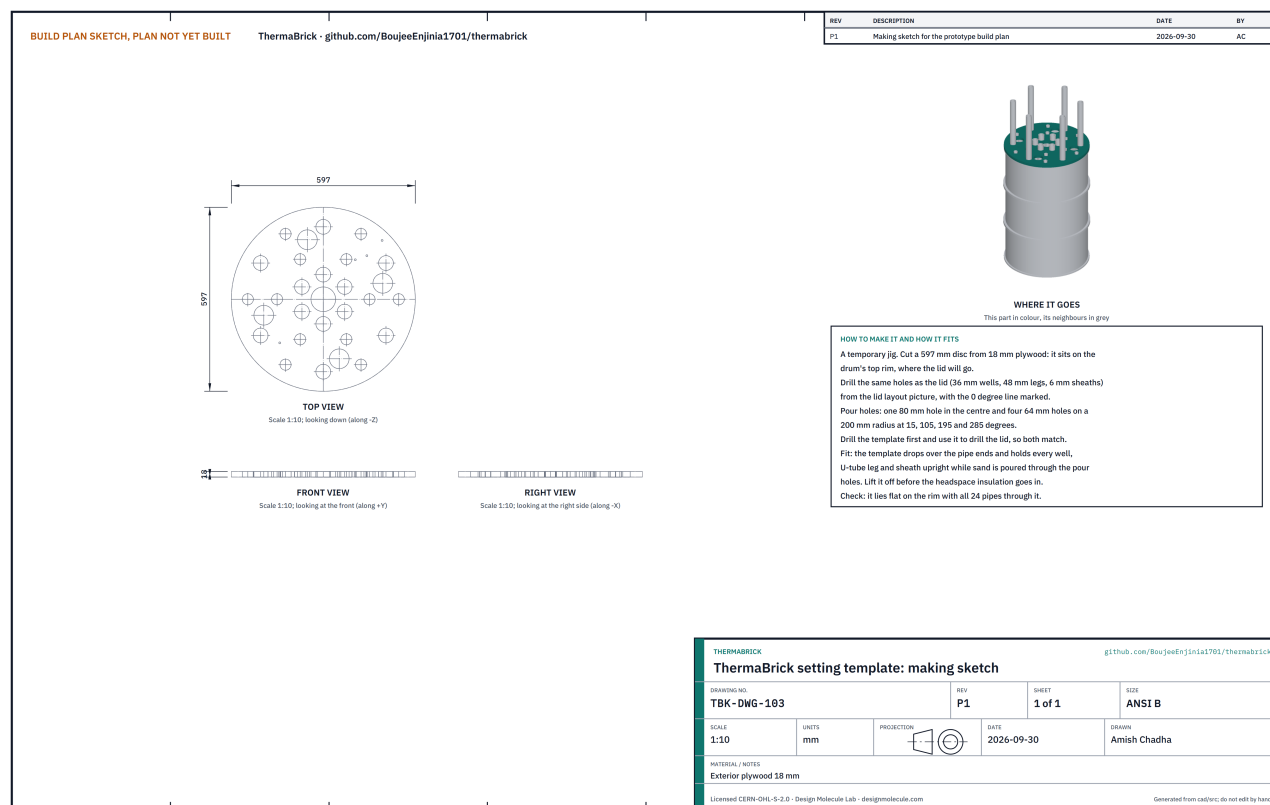


Figure 7. Setting template making sketch (TBK-DWG-103).

What it is and what it is made from. A temporary plywood disc that sits on the drum rim and holds every pipe and sensor upright while the sand is poured. Exterior plywood 18 mm, 610 x 610 mm.

How to make it.

1. Cut a 597 mm disc. Mark the centre and the 0° line.
2. Drill the 28 holes of Figure 6 with the same hole saws as the lid.
3. Cut the pour holes: 80 mm in the centre and four of 64 mm on a 200 mm radius at 15, 105, 195 and 285°.
4. Sand the edges so it lifts off without snagging.

How it fits the parts next to it. It lies on the drum's top rim where the lid will go, 0° marks together, clamped at four points. Every pour hole is at least 10 mm from a pipe hole.

Check before moving on. It lies flat on the rim.

3.4 Heater wells (make 12)

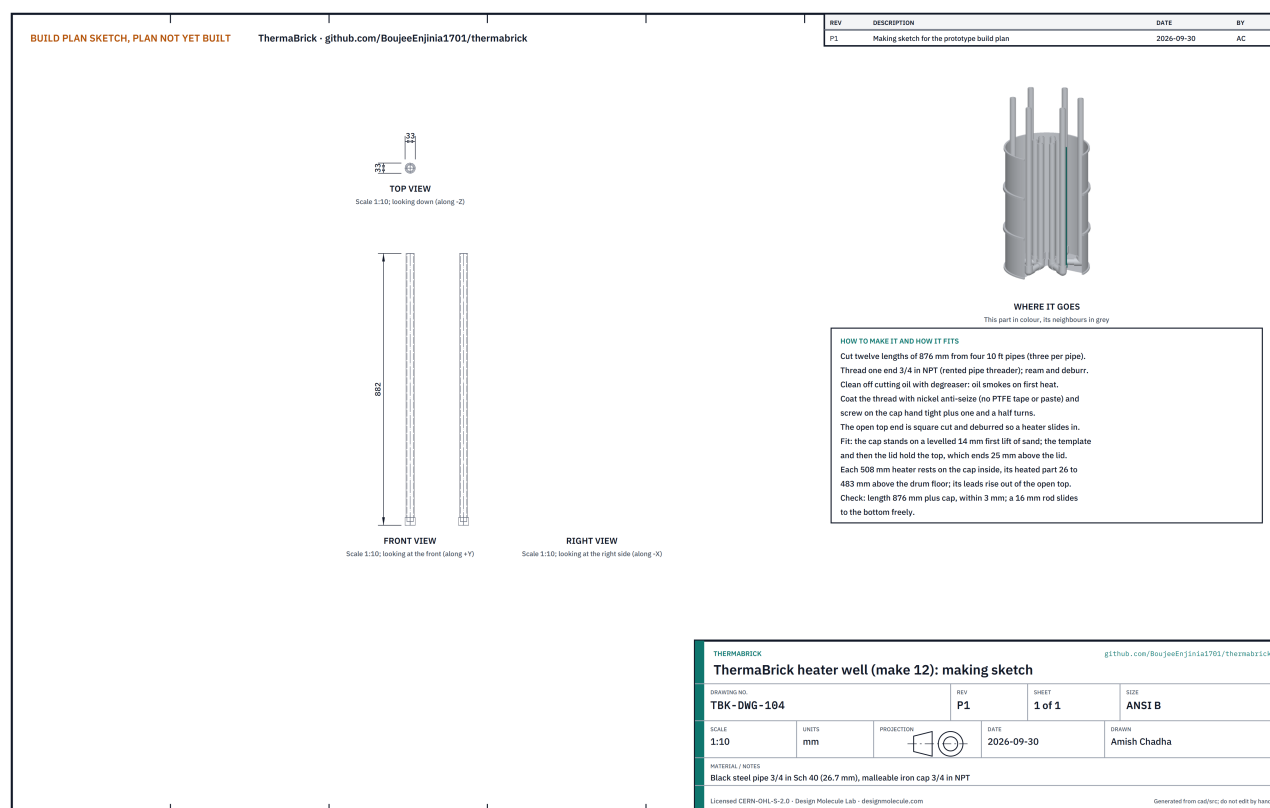


Figure 8. Heater well making sketch (TBK-DWG-104).

What it is and what it is made from. A capped steel pipe that a cartridge heater slides into, so the heater never touches the sand and can be pulled out from above. Black (uncoated) steel pipe 3/4 in Sch 40 (26.7 mm outside), four 10 ft lengths; twelve 3/4 in malleable iron caps.

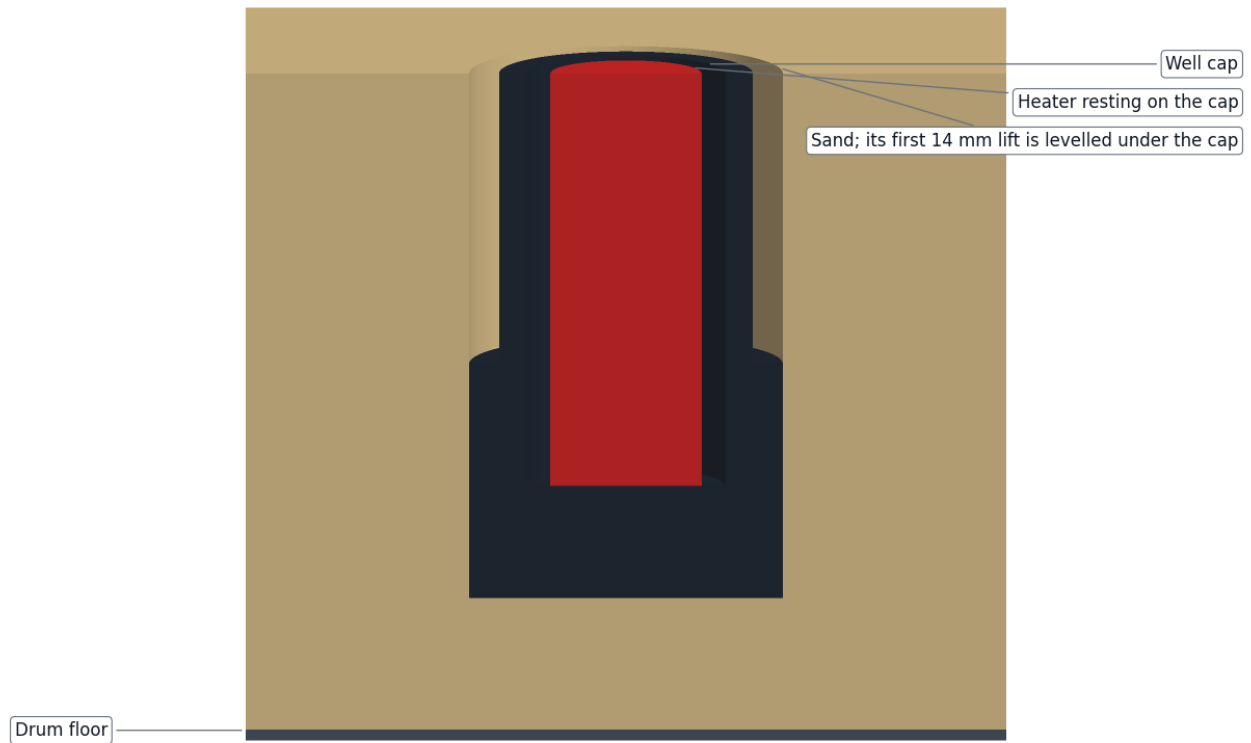
How to make it.

1. Cut twelve lengths of 876 mm, three from each pipe. Square the ends.
2. Thread one end of each 3/4 in NPT with a rented pipe threader; ream and deburr both ends.
3. Degrease inside and out: cutting oil smokes on the first firing.
4. Coat the thread with nickel anti-seize (never PTFE tape or paste) and screw on a cap, hand tight plus one and a half turns.

How it fits the parts next to it.

Joint 2: heater well foot (cut through the well)

The cap stands on a levelled 14 mm lift of sand, which keeps it off the drum floor; the heater rests in the cap



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Figure 9. The cap stands on a levelled 14 mm first layer of sand, which keeps it off the drum floor; the heater rests in the cap.

Each well stands on its cap at its mark, held upright by the template and then the lid; its open top ends 25 mm above the lid. The heater rests on the inside of the cap, its heated length from 26 to 483 mm above the drum floor, and its leads rise out of the open top.

Check before moving on. A 16 mm rod slides to the bottom of every well freely.

3.5 U-tubes (make 6)

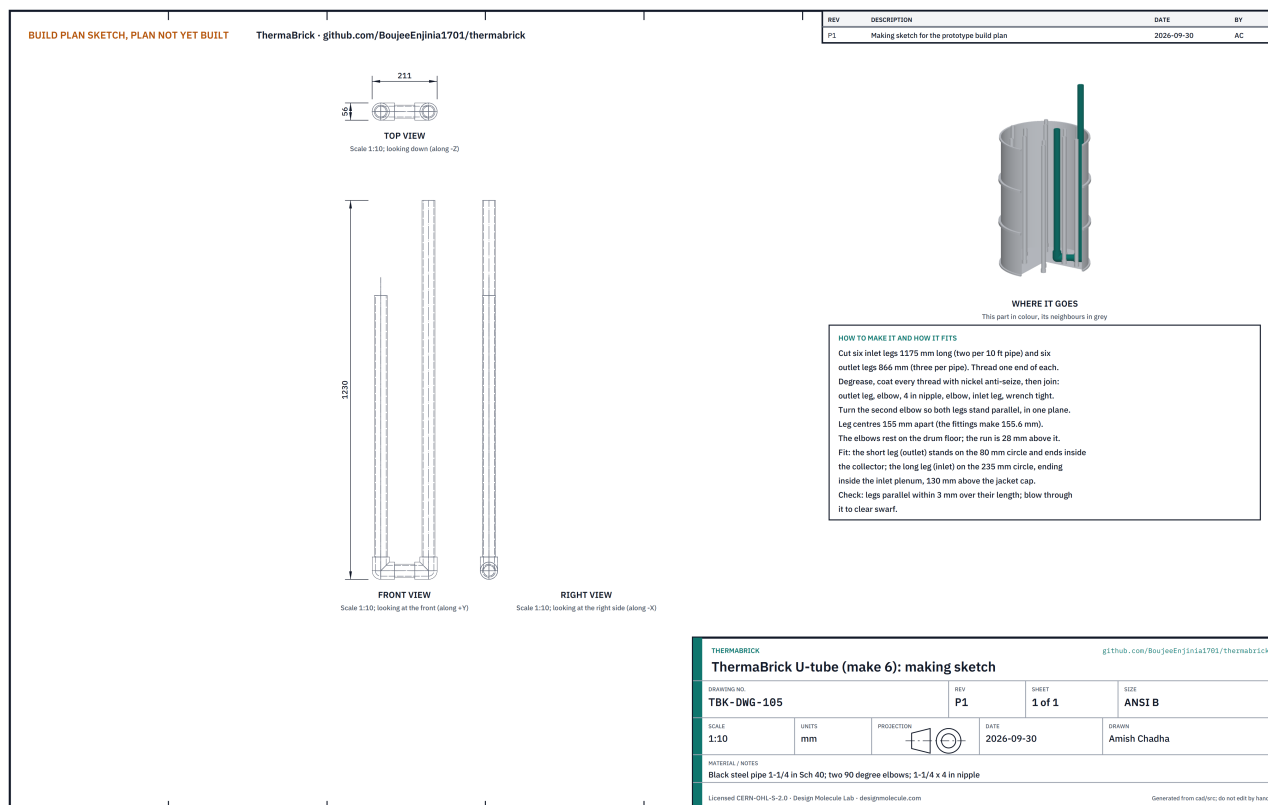


Figure 10. U-tube making sketch (TBK-DWG-105).

What it is and what it is made from. The heat exchanger: room air goes down the long leg, across the bottom and up the short leg, picking up heat from the sand without ever touching it. Black steel pipe 1-1/4 in Sch 40 (42.2 mm outside), five 10 ft lengths; twelve 1-1/4 in malleable iron 90° elbows; six 1-1/4 x 4 in black nipples.

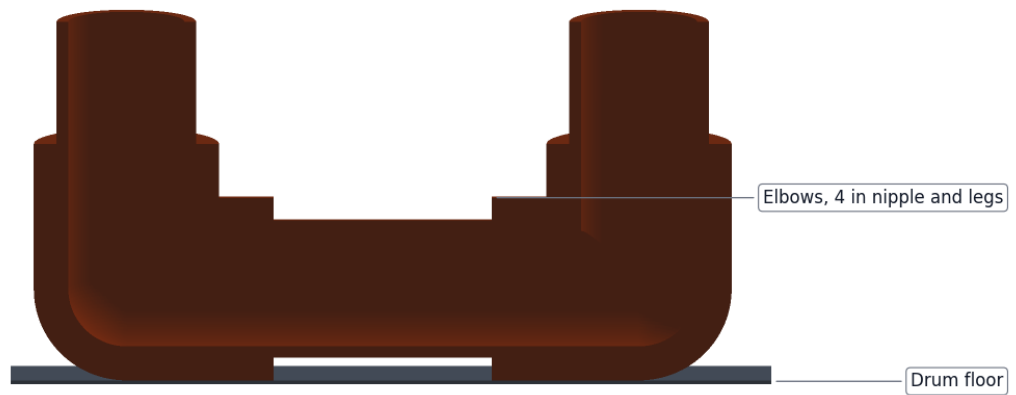
How to make it.

1. Cut six long (inlet) legs of 1,175 mm, two from each of three pipes, and six short (outlet) legs of 866 mm, three from each of two pipes.
2. Thread one end of each; ream, deburr and degrease.
3. Coat every thread with nickel anti-seize and join, wrench tight: short leg, elbow, 4 in nipple, elbow, long leg.
4. Turn the second elbow so the two legs are parallel and in one plane, 155 mm apart between centres.
5. Blow through each tube to clear swarf.

How it fits the parts next to it.

Joint 3: U-tube foot (cut through the tube)

Two elbows and a 4 in nipple set the legs 155 mm apart; the elbows rest on the floor



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Figure 11. Two elbows and a 4 in nipple set the legs 155 mm apart; the elbows rest on the drum floor.

The elbows rest on the drum floor, the bottom run 28 mm above it. The short leg stands on the 80 mm circle and ends 50 mm above the lid, inside the collector; the long leg stands on the 235 mm circle and ends inside the inlet plenum, 130 mm above the jacket cap.

Check before moving on. The legs are parallel within 3 mm over their length.

3.6 Thermocouples and guide rods

Joint 9: sand thermocouple on its guide rod

The rod stands on the floor 7 mm from the sheath; tie them every 100 mm with stainless wire



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Figure 12. Each sand thermocouple is tied to a stainless rod that stands on the drum floor.

What it is and what it is made from. Eight type K mineral-insulated thermocouples, 3 mm Inconel sheath, 1,500 mm long, with mini plugs; four stainless rods 4.8 mm, cut to 561 mm; stainless tie wire.

How to fit them. Their positions are in Figure 6.

1. Well walls: T1 on the inner-ring well at 0°, T2 on the outer-ring well at 180° (on its inner side), T3 on the inner-ring well at 120°. Tie each sheath to its well with stainless wire, its tip 255 mm above the drum floor (the middle of the heated length).
2. Sand: stand a guide rod on the floor 7 mm inboard of each sand position and tie the sheath to it every 100 mm. T5 (200 mm, 45°) and T6 (270 mm, 45°) with tips 286 mm above the floor; T7 (200 mm, 225°) with its tip 471 mm above the floor.
3. T4 measures the centre: its rod stands 7 mm off the centre, its tip 286 mm above the floor. Above the sand, bend its sheath once by hand and run it across the sand surface at 51°, between two short U-tube legs, to its lid hole at 165 mm.

- T8 goes in later, through the 4 mm hole in the collector top, into the collector air.
- Above the lid, every sheath runs in the top insulation to the exit grommet in the cap; coil the spare length flat in the insulation.

Check before moving on. Each thermocouple reads room temperature on a meter, and its tip is at its stated height within 10 mm.

3.7 Collector and outlet

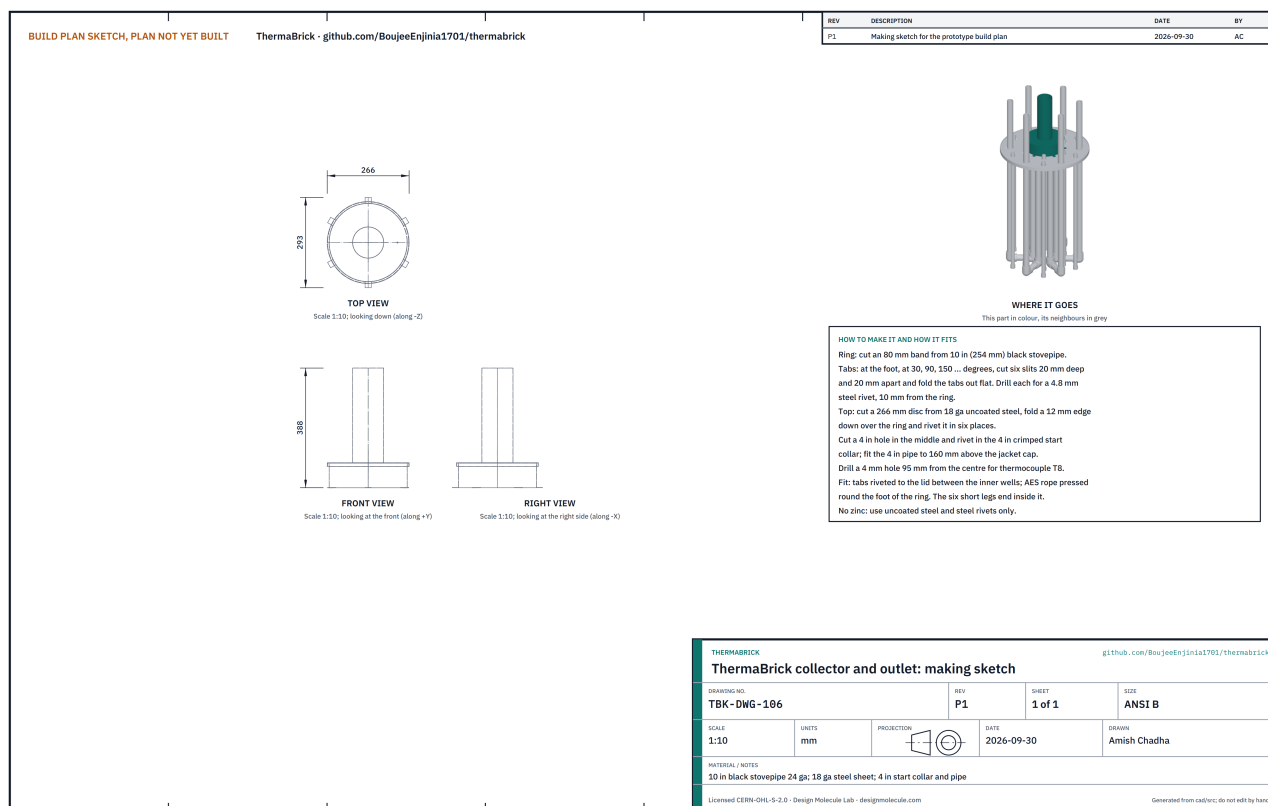


Figure 13. Collector and outlet making sketch (TBK-DWG-106).

What it is and what it is made from. A shallow drum on the lid that gathers the hot air from the six short legs and sends it up the 4 in outlet. A section of 10 in (254 mm) black stovepipe, 24 ga; 18 ga uncoated steel sheet; a 4 in crimped start collar and 4 in black steel pipe; steel pop rivets.

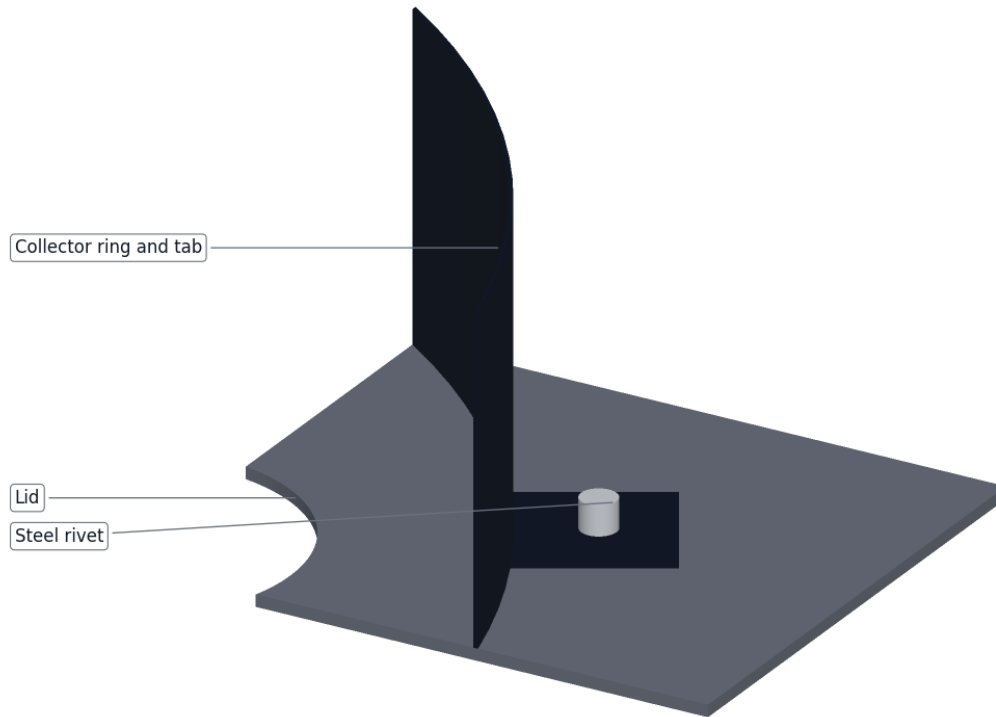
How to make it.

- Cut an 80 mm band from the stovepipe and close its seam.
- At the foot, at 30, 90, 150, 210, 270 and 330°, make two 20 mm slits 20 mm apart and fold each tab out flat. Drill each tab 4.9 mm, 10 mm out from the ring.
- Cut a 266 mm disc of 18 ga sheet; fold a 12 mm edge down and rivet it over the ring in six places.
- Cut a 4 in hole in the middle of the disc and rivet in the start collar. Drill a 4 mm hole 95 mm from the centre for T8.
- Fit the 4 in pipe in the collar, long enough to stand 160 mm above the jacket cap (about 300 mm).

How it fits the parts next to it.

Joint 5: collector tab riveted to the lid

Six tabs folded out at the foot of the ring, between the inner wells; one steel rivet each



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Figure 14. Each tab lies on the lid between two inner-ring wells and is held by one steel rivet.

The ring stands on the lid over the six short legs, 9.6 mm clear of the inner wells. Each tab is riveted to the lid with a steel rivet (never aluminium in the hot zone), and AES rope is pressed round the foot of the ring.

Check before moving on. The collector sits flat; no tab touches a well.

3.8 Jacket side

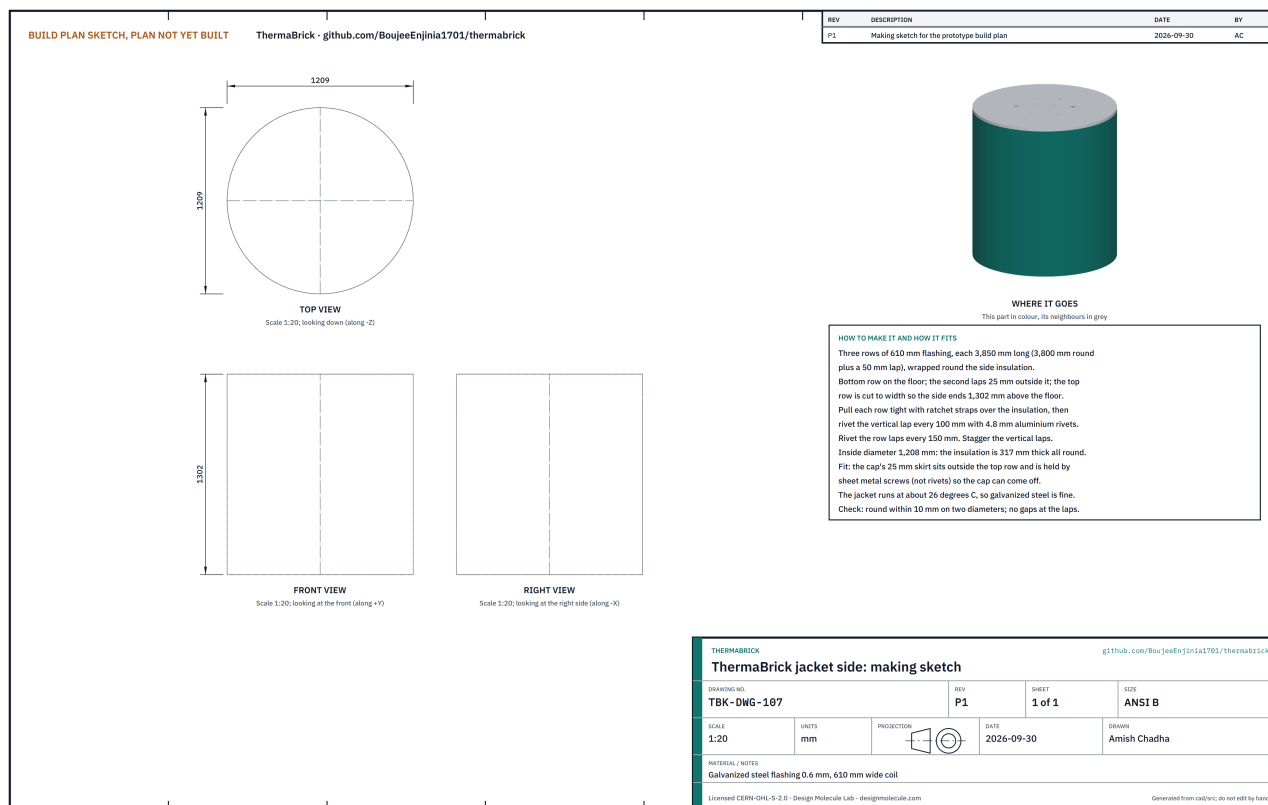


Figure 15. Jacket side making sketch (TBK-DWG-107).

What it is and what it is made from. The outer skin, 1,210 mm across and 1,302 mm high, that holds the side insulation. Galvanized steel flashing 0.6 mm, 610 mm wide coil; aluminium pop rivets.

How to make it.

1. Cut three strips 3,850 mm long (3,800 mm round plus a 50 mm lap). Cut the top strip to the width that brings the side to 1,302 mm with 25 mm laps between rows.
2. Wrap the bottom strip round the insulated unit on the floor, pull it tight with two ratchet straps, and rivet the vertical lap every 100 mm.
3. Add the second and top rows, each lapping 25 mm outside the one below, the vertical laps staggered. Rivet the row laps every 150 mm.

How it fits the parts next to it. The jacket stands on the slab round the base and holds the 317 mm of side insulation. The cap's skirt sits outside its top edge (Figure 18). The jacket runs at about 26 °C, so galvanized steel is allowed here.

Check before moving on. Two diameters at right angles agree within 10 mm; no gap at any lap.

3.9 Jacket cap

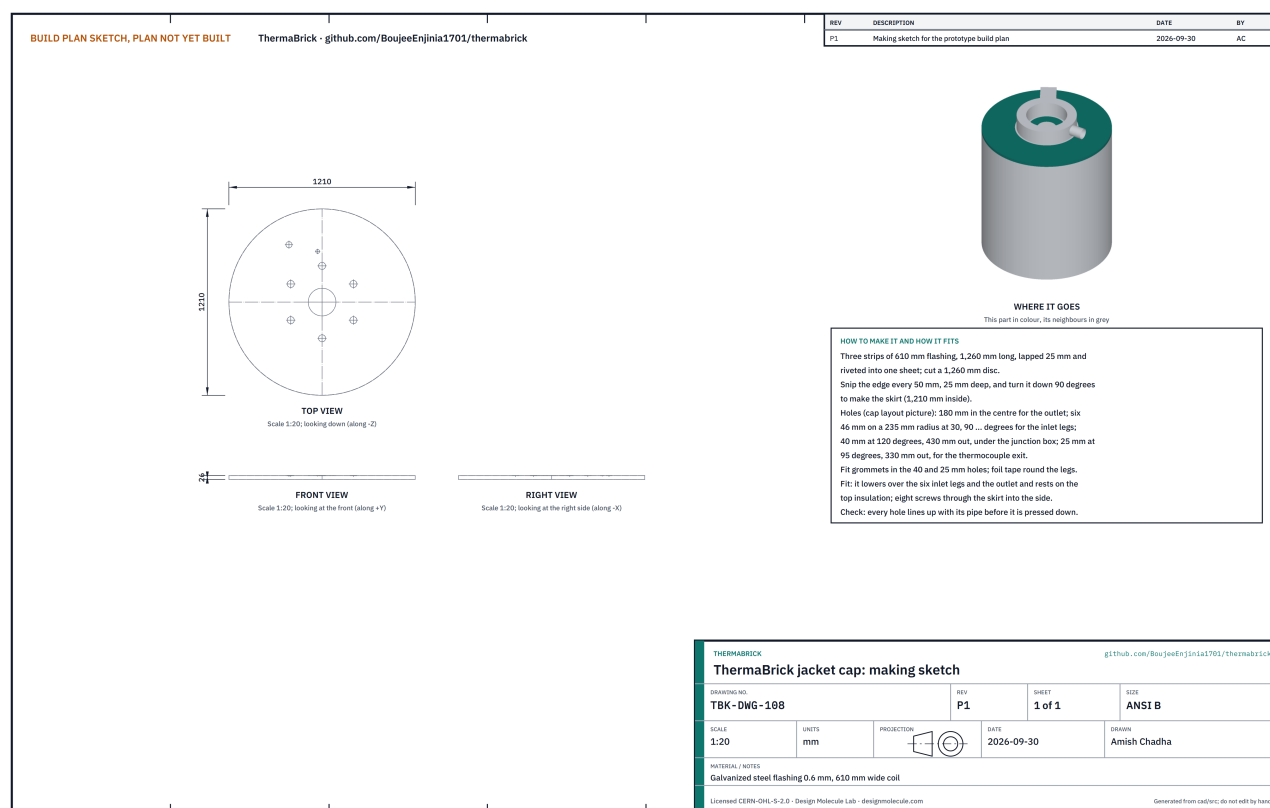
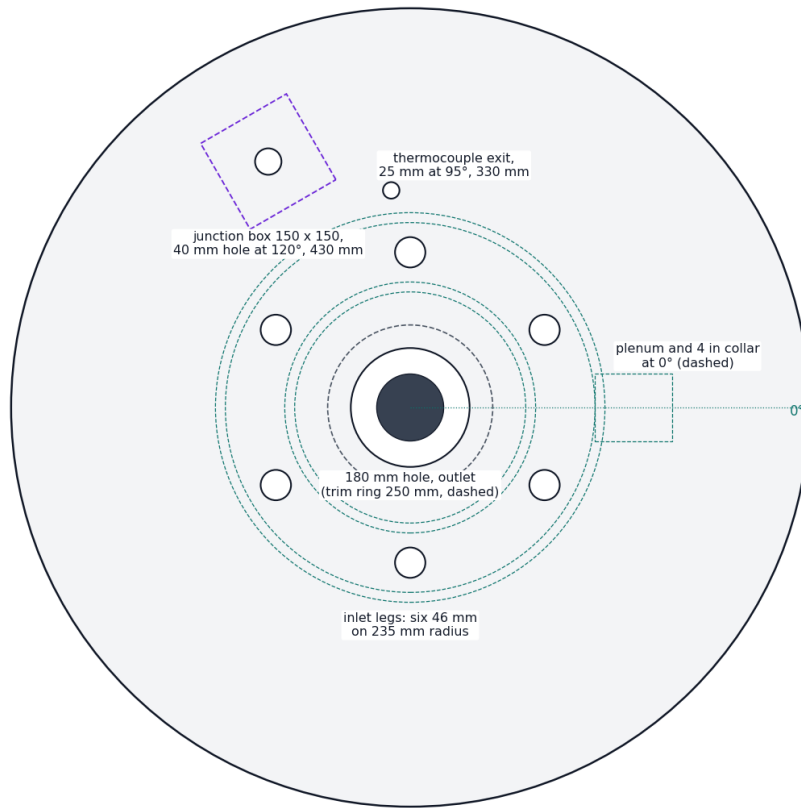


Figure 16. Jacket cap making sketch (TBK-DWG-108).

Jacket cap: hole layout and what sits on it

Seen from above. Solid: holes to cut. Dashed: outlines of the parts that sit on the cap.



Cap holes and parts

Cap 1,260 mm blank, skirt turned down
25 mm (1,210 mm inside)

Holes:

180 mm centre (outlet, packed with AES, covered by the black steel trim ring)
six 46 mm on 235 mm radius at 30, 90 ...
40 mm at 120°, 430 mm out (heater leads, grommet, under the junction box)
25 mm at 95°, 330 mm out (thermocouple exit, grommet)

Sits on the cap:

inlet plenum, 380 to 560 mm across, flanges 350 to 590 mm
junction box at 120°, 430 mm out
trim ring over the centre hole

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Figure 17. Jacket cap hole layout and the outlines of the parts that sit on it.

What it is and what it is made from. The removable top of the jacket. Three strips of the same flashing, stainless sheet metal screws, two silicone grommets.

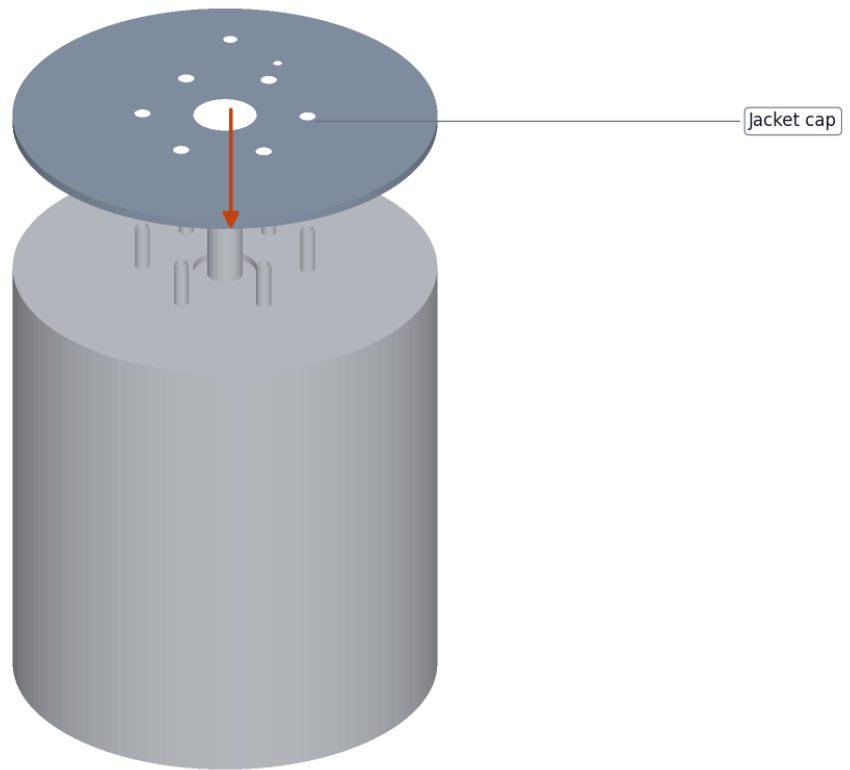
How to make it.

1. Cut three strips 1,260 mm long, lap them 25 mm and rivet the laps every 100 mm. Cut a 1,260 mm disc.
2. Snip the edge every 50 mm, 25 mm deep, and turn it down 90° to make the skirt.
3. Cut the holes of Figure 17: 180 mm in the centre for the outlet; six of 46 mm on a 235 mm radius at 30, 90, 150, 210, 270 and 330° for the long legs; 40 mm at 120°, 430 mm out, under the junction box; 25 mm at 95°, 330 mm out, for the thermocouple exit.
4. Fit grommets in the 40 mm and 25 mm holes.

How it fits the parts next to it. It lowers over the six long legs and the outlet and rests on the top insulation; the heater leads and thermocouple sheaths come up through the grommets. Eight sheet metal screws through the skirt hold it to the jacket side, so it can be lifted off to reach a heater.

Step 16: jacket cap

Over the inlet legs and outlet; leads and sheaths through their grommets; eight screws in the skirt



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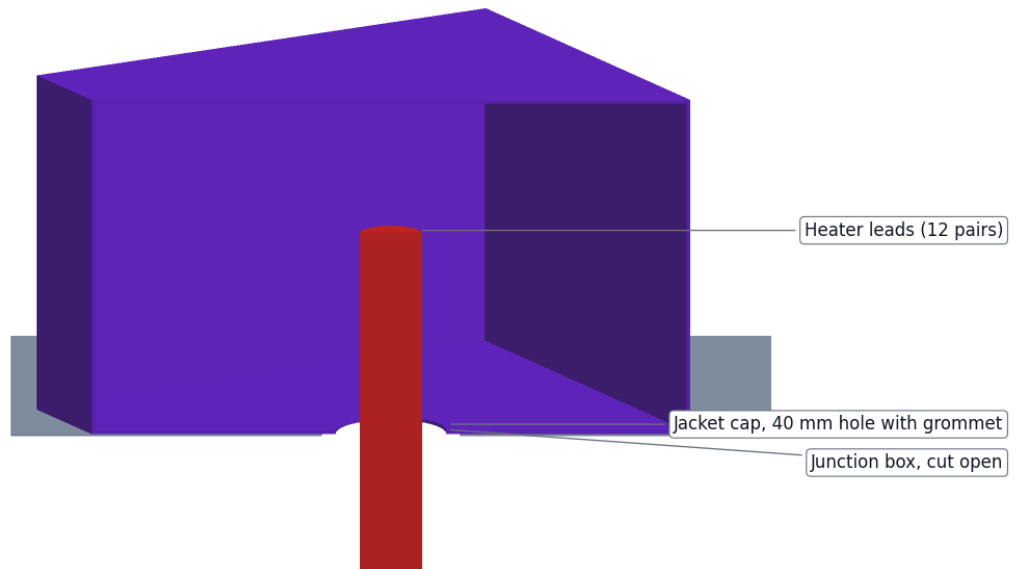
Figure 18. The cap lowers over the six long legs and the outlet; its skirt sits outside the top row of the jacket side.

Check before moving on. Every hole lines up with its pipe before the cap is pressed down.

3.10 Heater junction box

Joint 8: junction box on the cap (cut)

The heater leads rise through the top insulation and a grommet into the box; four screws hold it



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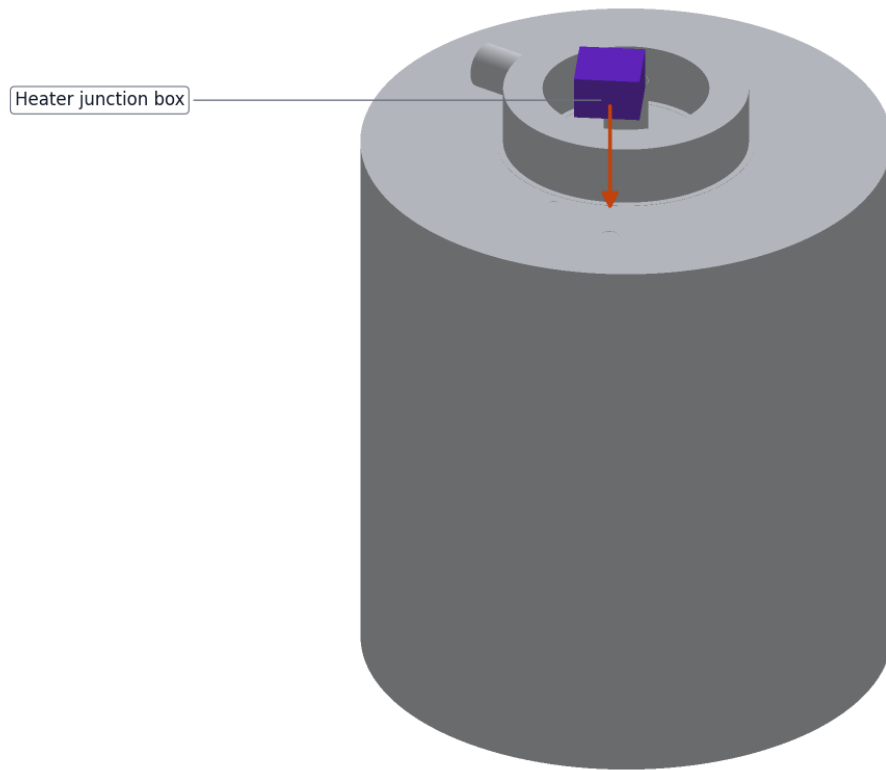
Figure 19. The heater leads rise through the top insulation and the grommet into the box.

What it is. A bought steel box, 150 x 150 x 100 mm, with ceramic terminal blocks, where the 24 heater leads meet the cable from the control enclosure.

What to do to it. Drill a 40 mm hole in its floor to match the cap hole, and four 4.5 mm holes for screws. Fit a cable gland for the supply cable. It sits on the cap at 120°, 430 mm out, 60 mm clear of the plenum.

Step 19: heater junction box

Seen from behind. Leads through the grommet into the box; land them on the ceramic terminal blocks; four screws



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Figure 20. The box goes on last, over the leads (assembly step 19).

3.11 Outlet trim ring

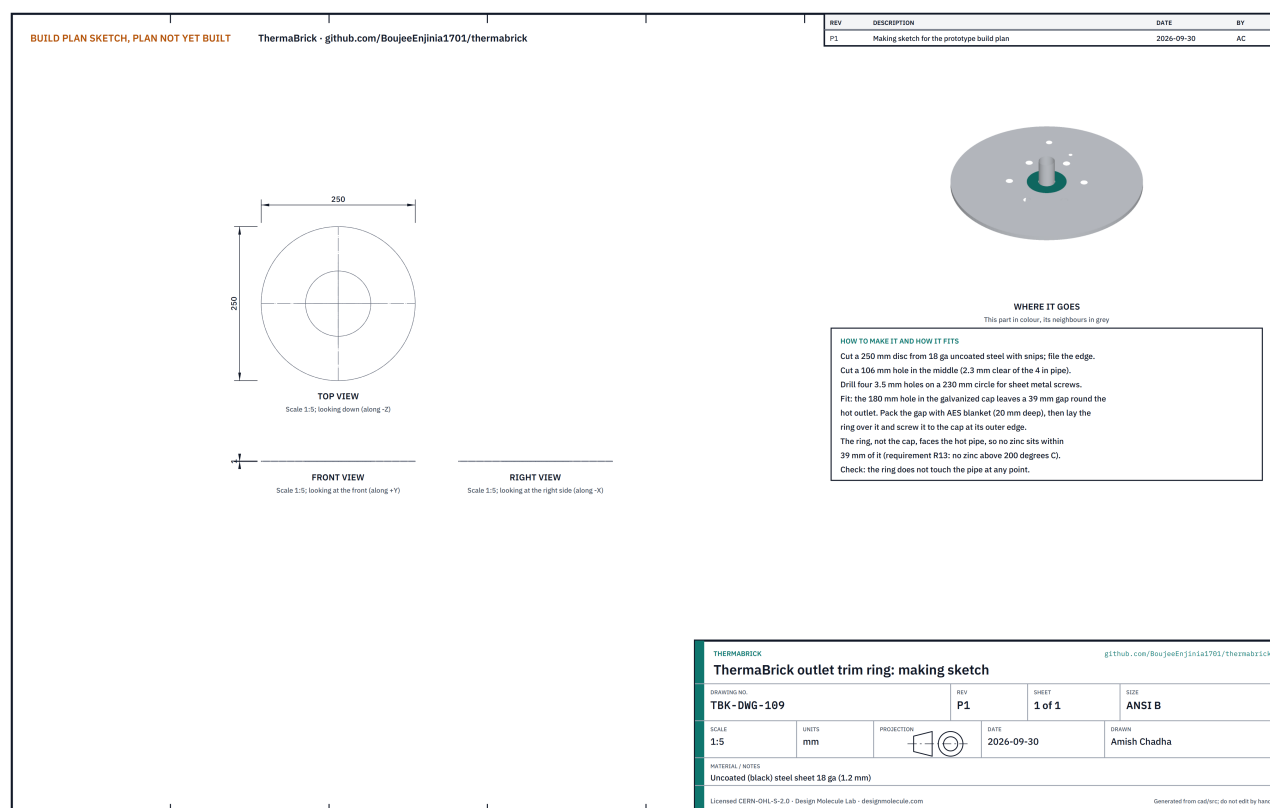


Figure 21. Outlet trim ring making sketch (TBK-DWG-109).

What it is and what it is made from. A flat steel ring that covers the fiber packing where the hot outlet passes through the cap. 18 ga (1.2 mm) uncoated steel sheet.

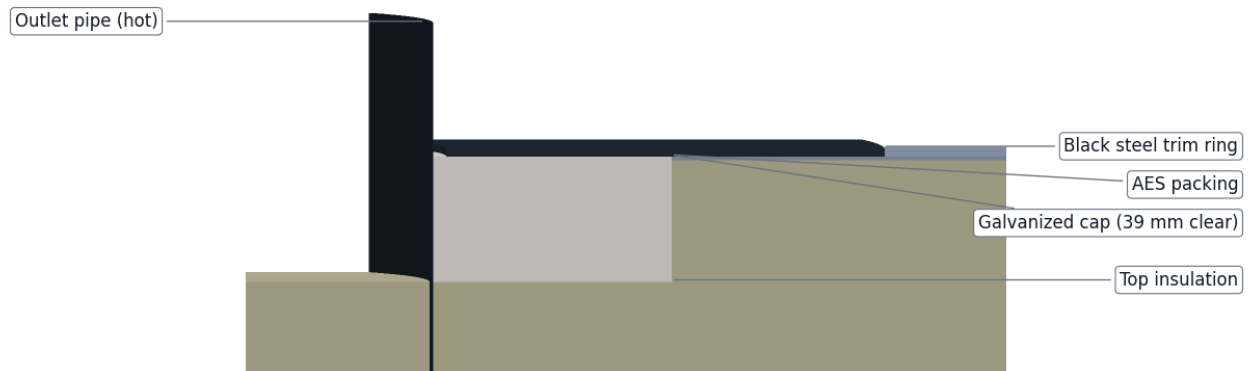
How to make it.

1. Cut a 250 mm disc with snips and file the edge.
2. Cut a 106 mm hole in the middle.
3. Drill four 3.5 mm holes on a 230 mm circle.

How it fits the parts next to it.

Joint 6: hot outlet through the jacket cap (cut)

Section, seen from the front. The galvanized cap stops 39 mm from the hot pipe; AES packing fills the gap; the trim ring covers it



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Figure 22. The galvanized cap stops 39 mm from the hot pipe; fiber packing fills the gap and the trim ring covers it.

The 39 mm gap between the outlet and the cap is packed 20 mm deep with AES blanket. The ring lies on the packing and the cap, 2.3 mm clear of the pipe all round, held by four screws at its outer edge.

Check before moving on. The ring touches the pipe nowhere.

3.12 Inlet plenum

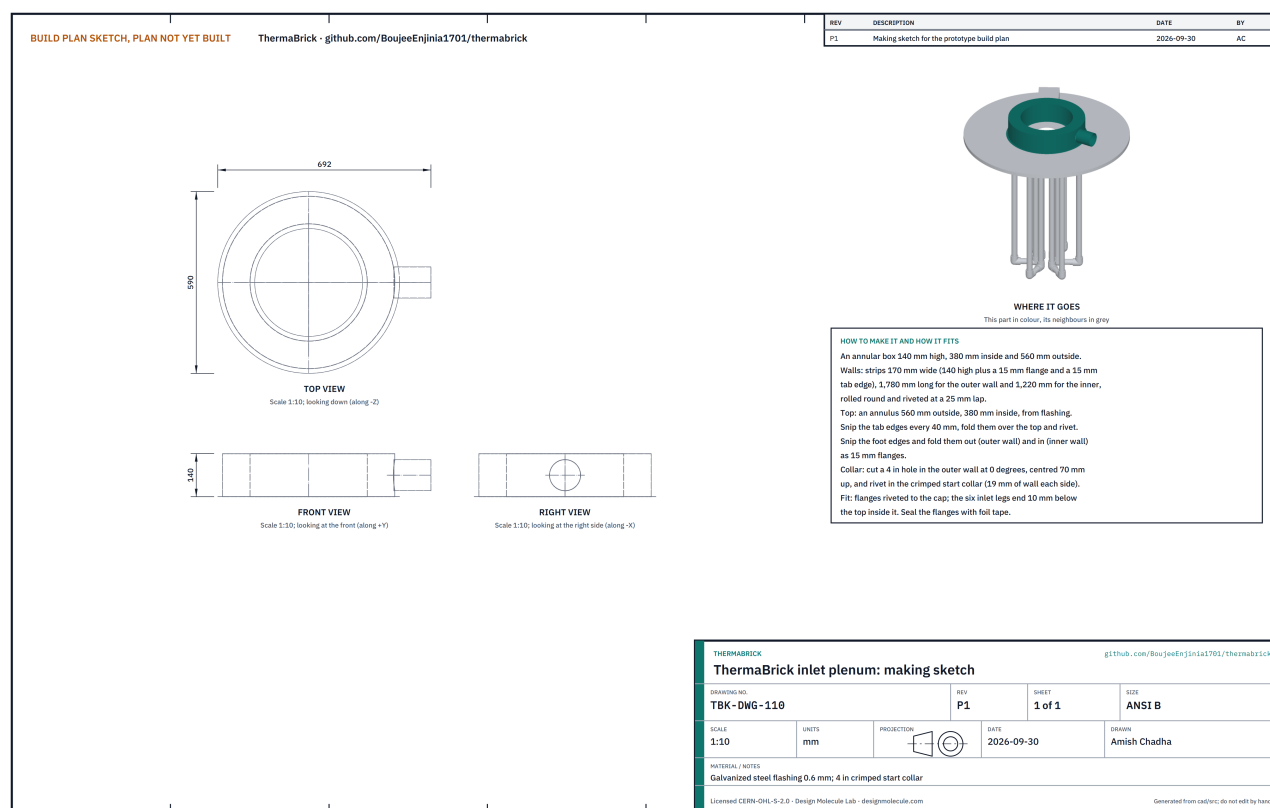


Figure 23. Inlet plenum making sketch (TBK-DWG-110).

What it is and what it is made from. A ring-shaped sheet metal box on the cap that takes room air from the 4 in inlet collar and shares it among the six long legs. Galvanized flashing 0.6 mm; a 4 in crimped start collar; pop rivets; foil tape.

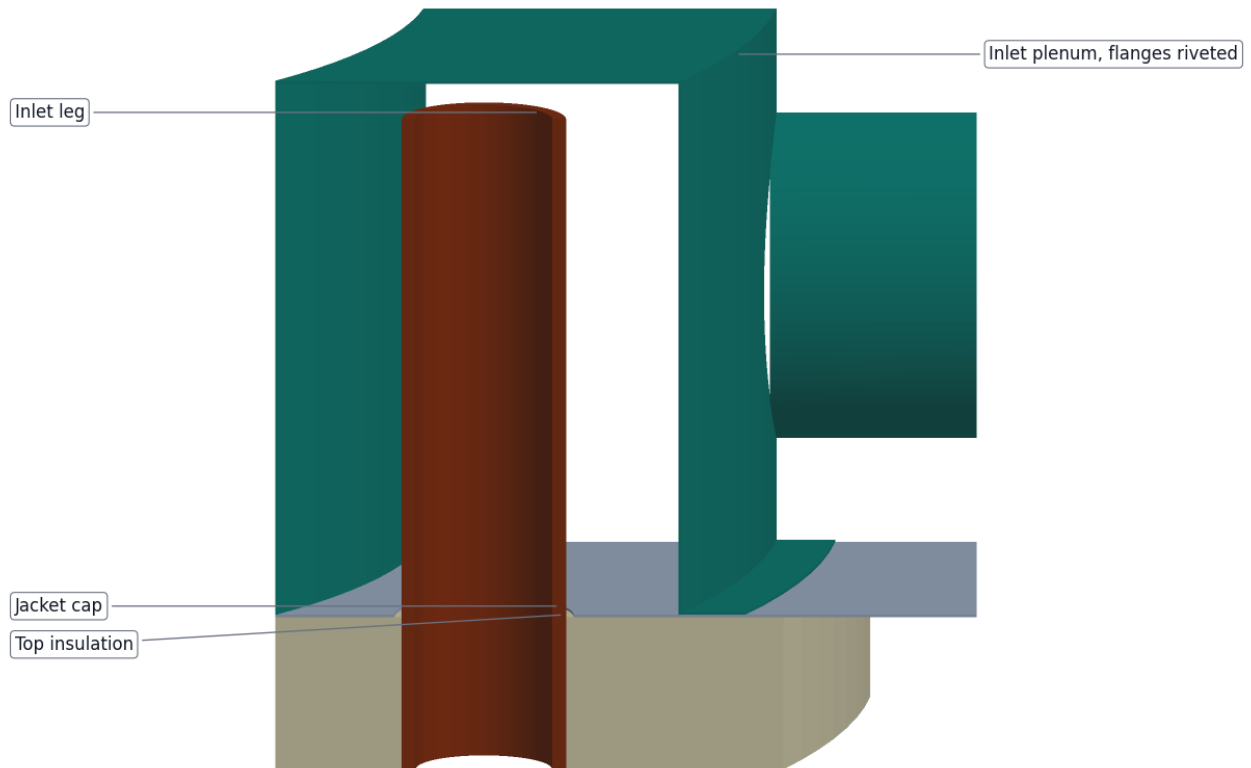
How to make it.

1. Cut two strips 170 mm wide: 1,780 mm long for the outer wall and 1,220 mm for the inner wall. Roll each into a ring (560 mm and 380 mm across) and rivet a 25 mm lap.
2. Cut the top: a ring 560 mm outside and 380 mm inside.
3. On each wall, snip the top 15 mm every 40 mm, fold the tabs over the top and rivet them.
4. Snip the bottom 15 mm every 40 mm and fold the tabs out (outer wall) or in (inner wall) as flanges.
5. In the outer wall at 0°, cut a 4 in hole centred 70 mm up and rivet in the start collar, which leaves 19 mm of wall above and below it.

How it fits the parts next to it.

Joint 7: inlet leg into the plenum (cut through the leg)

The leg passes the cap and ends 10 mm below the plenum top; the flanges sit on the cap



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Figure 24. A long leg passes through the cap and ends 10 mm below the plenum top.

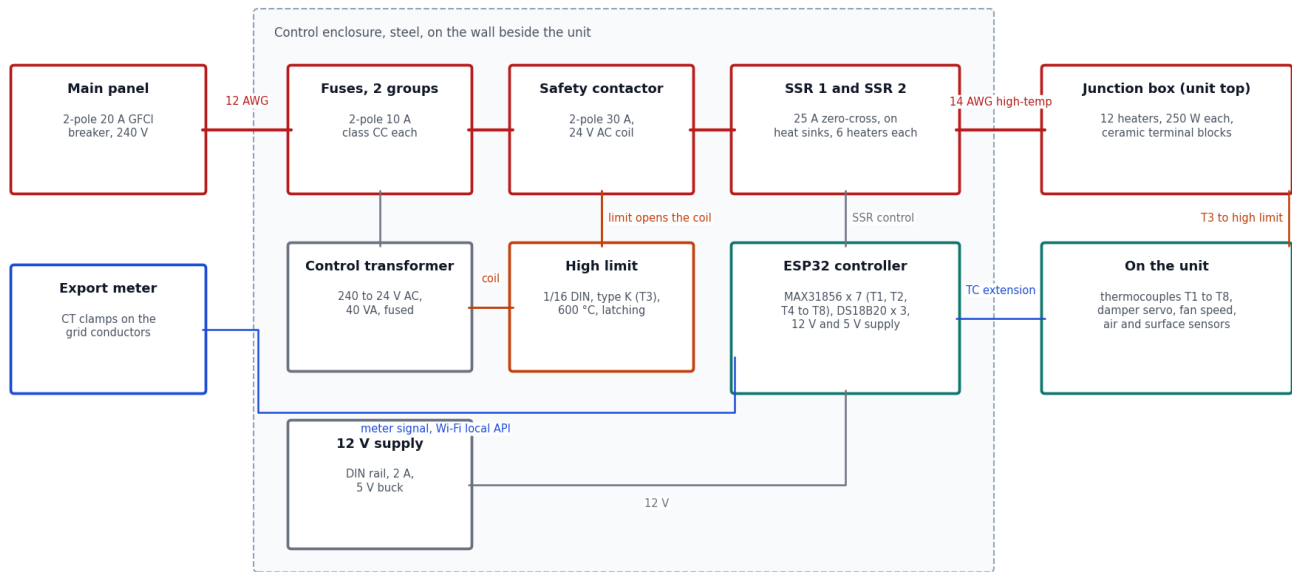
The plenum sits over the six long legs with its flanges on the cap, riveted and sealed with foil tape. The inlet damper fits on the collar.

Check before moving on. Air blown into the collar comes out of the outlet; none escapes at the flanges.

3.13 Wiring

ThermaBrick prototype: block-level wiring and the safety chain

240 V work is done or checked by a licensed electrician. Bought modules, wired at block level; no circuit board is laid out.



Safety chain: the high limit (T3 at 600 °C) opens the 24 V coil, and the contactor opens both legs of both heater groups. It latches until reset by hand.

Red: 240 V power. Blue: signal. Orange: safety chain. Grey: low voltage control. Heater leads to the junction box are the heaters' own 72 in ceramic-beaded leads.

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Figure 25. Block-level wiring. The 240 V circuit and the control enclosure are wired or checked by a licensed electrician.

The heaters are wired as two groups of six (1.5 kW each), each through a 10 A fuse pair and a solid-state relay, with a 2-pole safety contactor ahead of both groups. The contactor's 24 V coil runs through the independent high limit on thermocouple T3, which opens it at 600 °C and latches until reset by hand. The ESP32 controller reads T1, T2 and T4 to T8, the air and surface sensors and the export meter, and drives the relays, the damper servo and the fan speed. Wire to the bill of materials: 12 AWG to the enclosure on a dedicated 20 A circuit with 2-pole ground-fault protection; high-temperature 14 AWG from the enclosure to the junction box; type K extension cable from the exit grommet to the amplifiers.

3.14 Bought components

Buy to specification, not brand. Line numbers are those of the bill of materials.

- **Drum (line 1).** New, unlined, open-head 55 US gal steel drum, 18 ga body, 16 ga lid with bolted ring, 571.5 mm inside.
- **Sand (line 2).** Ten 50 lb (22.7 kg) bags of washed play sand, sieved to remove fines.
- **Heaters (line 3).** Twelve cartridge heaters 5/8 x 20 in (15.9 x 508 mm), 250 W at 240 V, Incoloy 800 sheath, 2 in cold end, **72 in (1,829 mm) ceramic-beaded leads.**
- **Pipe and fittings (lines 4 to 9).** As sections 3.4 and 3.5, with nickel anti-seize.
- **Collector parts (lines 10, 11).** As section 3.7.

- **Air path (lines 12 to 18).** 4 in black stovepipe kit with two adjustable elbows, 4 in tee, 4 in galvanized butterfly damper with a hobby servo, 4 in balancing damper, 4 in EC inline fan (60 L/s at 150 Pa, speed input, rated for 60 °C air), insulated flexible duct and a register.
- **Insulation (lines 19 to 24).** Two rolls of 25 mm AES blanket, one AES board, 16 firebricks, three packs of 89 mm stone wool batt, four 50 mm stone wool boards (60 kPa or more at 10 % strain), two 3 m lengths of 12 mm AES rope.
- **Jacket (lines 25, 26).** Two coils of 610 mm galvanized flashing; aluminium and steel pop rivets, stainless sheet metal screws, foil tape.
- **Controls (lines 27 to 42).** As Figure 25 and the bill of materials.
- **New for construction (lines 43 to 46).** Plywood for the template, 18 ga uncoated steel for the trim ring, four stainless guide rods, silicone grommets.

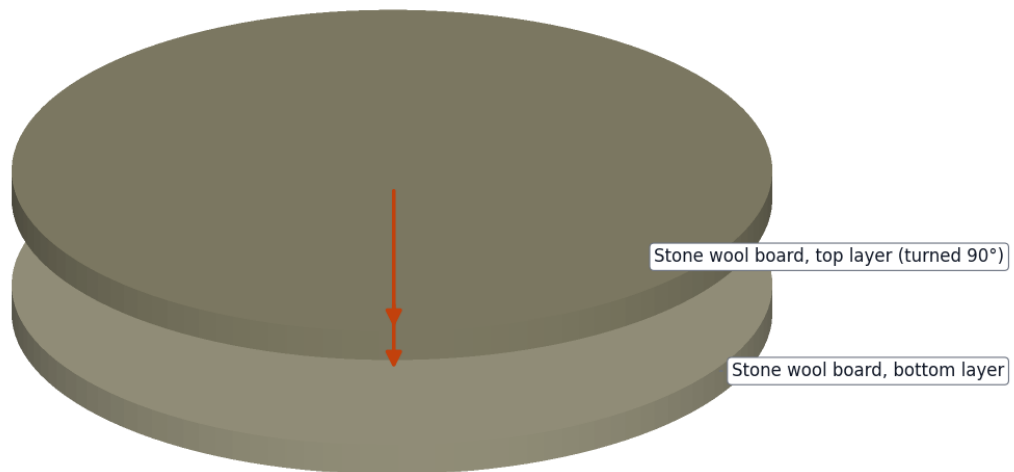
■ 4. Putting it together

In each picture the parts already fitted are grey and the part being fitted is in colour, with an arrow showing the way it goes in. The drum is drawn with a quarter cut away where the inside matters.

Step 1: stone wool boards on the slab

Step 1: stone wool boards on the slab

Two layers, joints crossed, on a clean, level concrete slab

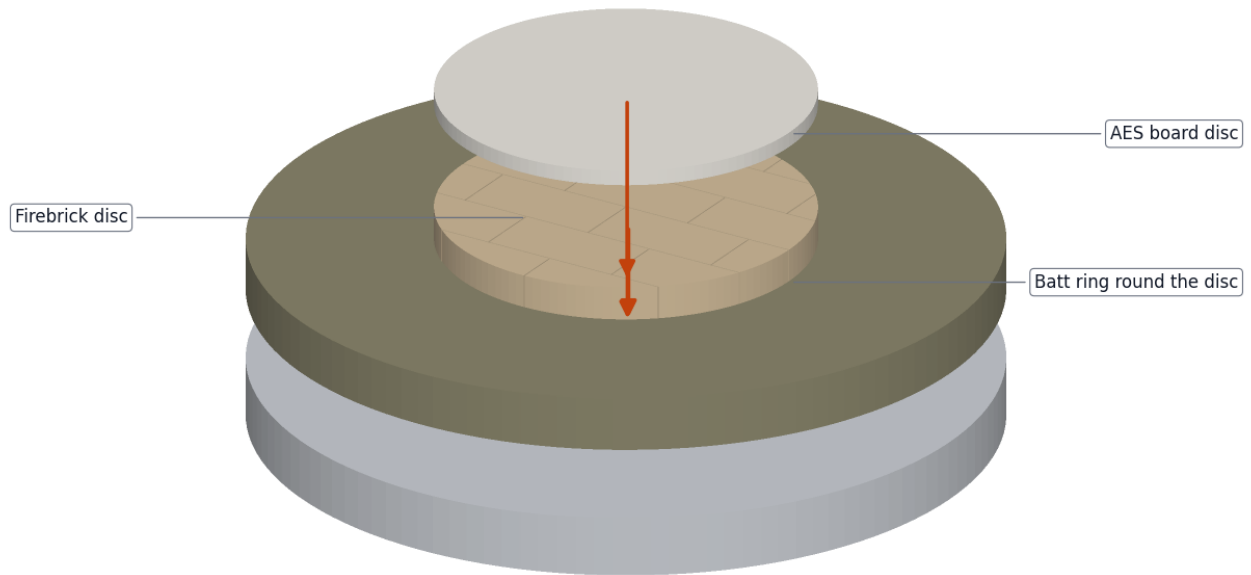


On a clean, level concrete slab inside the heated space, at least 450 mm from anything combustible. Two layers, joints crossed.

Step 2: firebrick disc, AES disc and batt ring

Step 2: firebrick disc, AES disc and batt ring

Bricks to the cutting layout; AES board disc on top; batt segments round them, level



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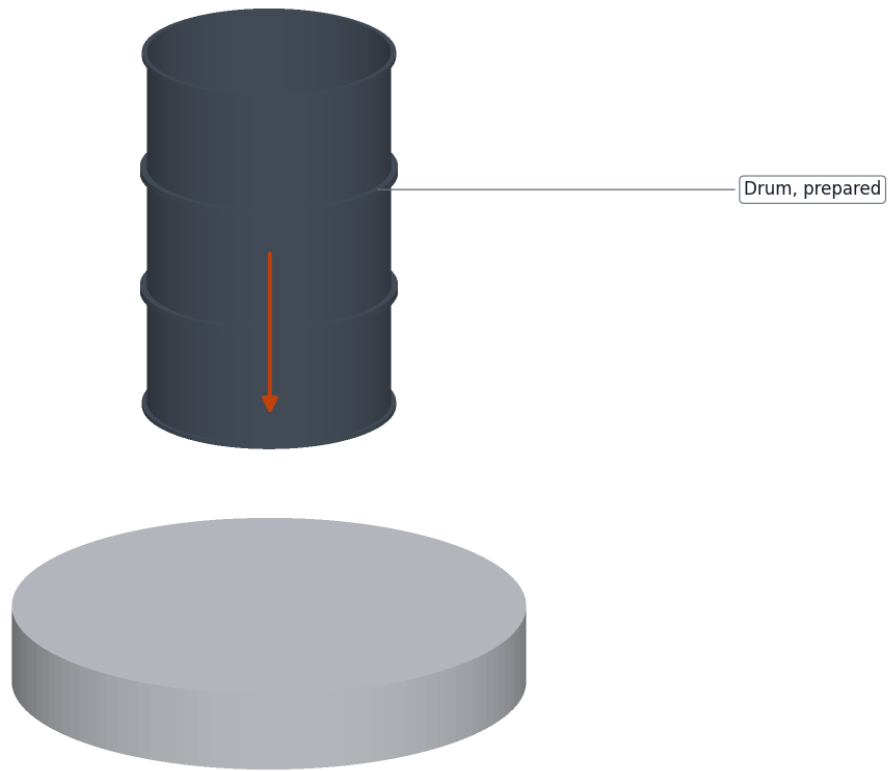
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Bricks to the cutting layout, dry, in the middle; the AES disc on top; the batt ring round both, level with the AES disc.

Step 3: drum onto the base

Step 3: drum onto the base

Empty and prepared; centred on the AES disc, 0° mark on the rim facing the inlet side



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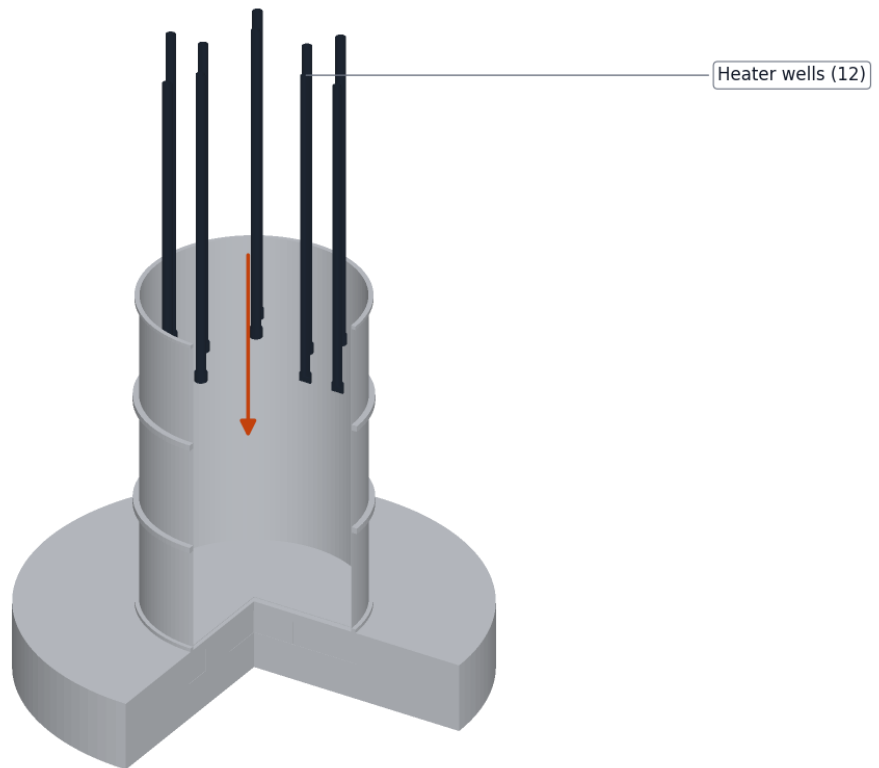
github.com/BoujeeEnjinia1701/thermabrick

Empty and prepared, centred on the AES disc (two people lift it), with its 0° mark facing where the inlet duct will come from. It is not moved again.

Step 4: heater wells into the drum

Step 4: heater wells into the drum

Level a 14 mm first lift of sand; stand each capped well on it at its mark; tape over the open tops



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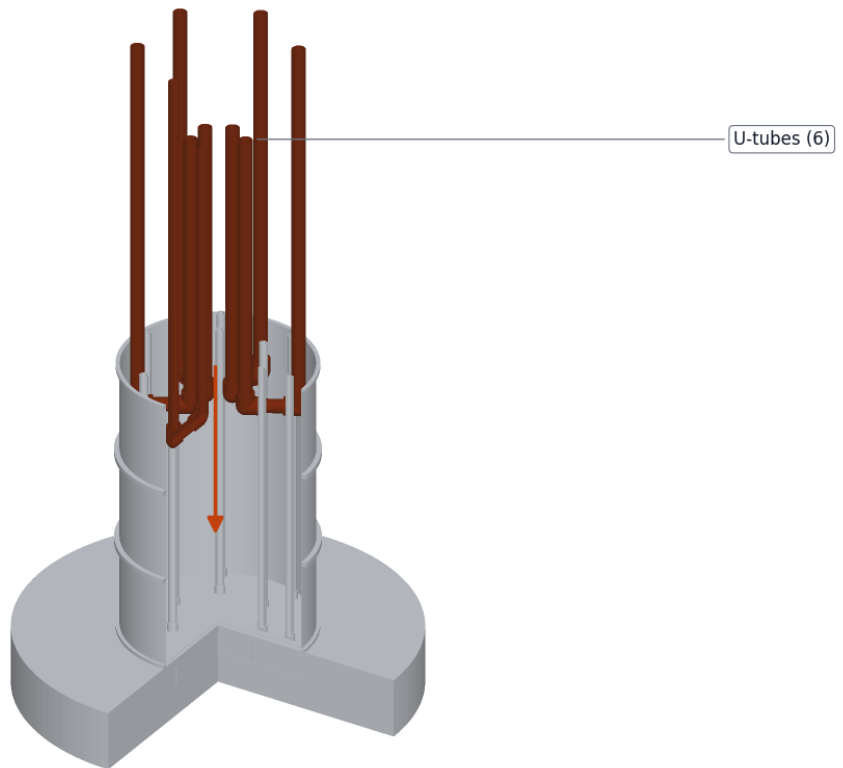
github.com/BoujeeEnjinia1701/thermabrick

Pour and level a 14 mm first layer of sand with a short board. Stand each capped well on it at its mark from Figure 6. Tape over the open tops so no sand gets in.

Step 5: U-tubes into the drum

Step 5: U-tubes into the drum

Elbows on the floor, short legs on the 80 mm circle, long legs on the 235 mm circle; tape the tops



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

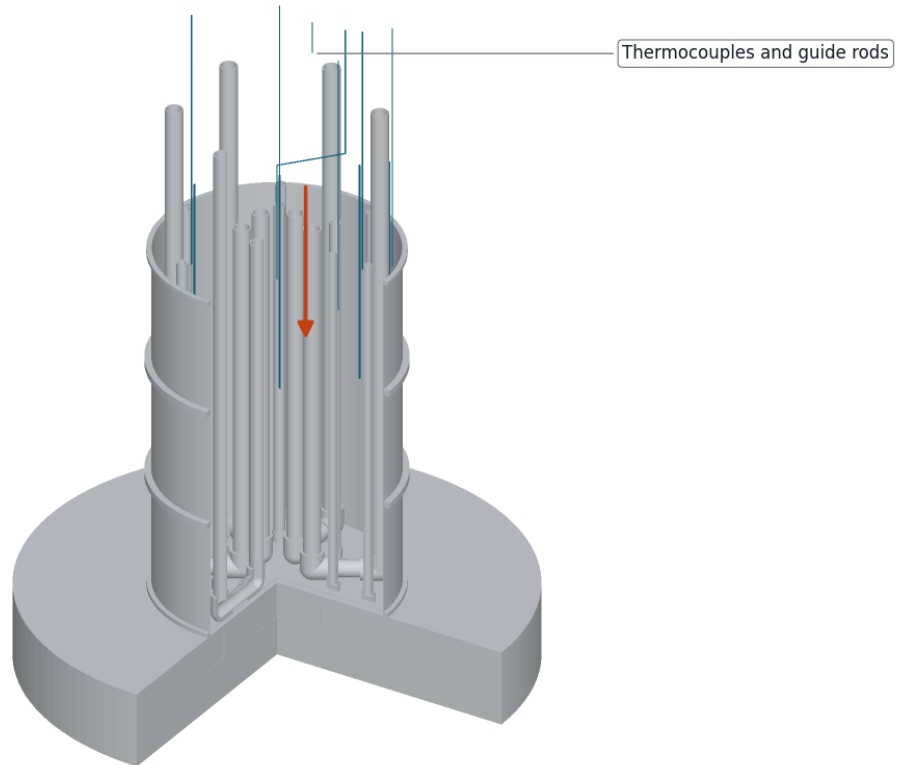
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Elbows on the floor (scrape the sand clear under them), short legs on the 80 mm circle, long legs on the 235 mm circle, each tube on its ray at 30, 90, 150 ... °. Tape the tops.

Step 6: thermocouples and guide rods

Step 6: thermocouples and guide rods

T1 to T3 wired to their wells; T4 to T7 tied to guide rods standing on the floor



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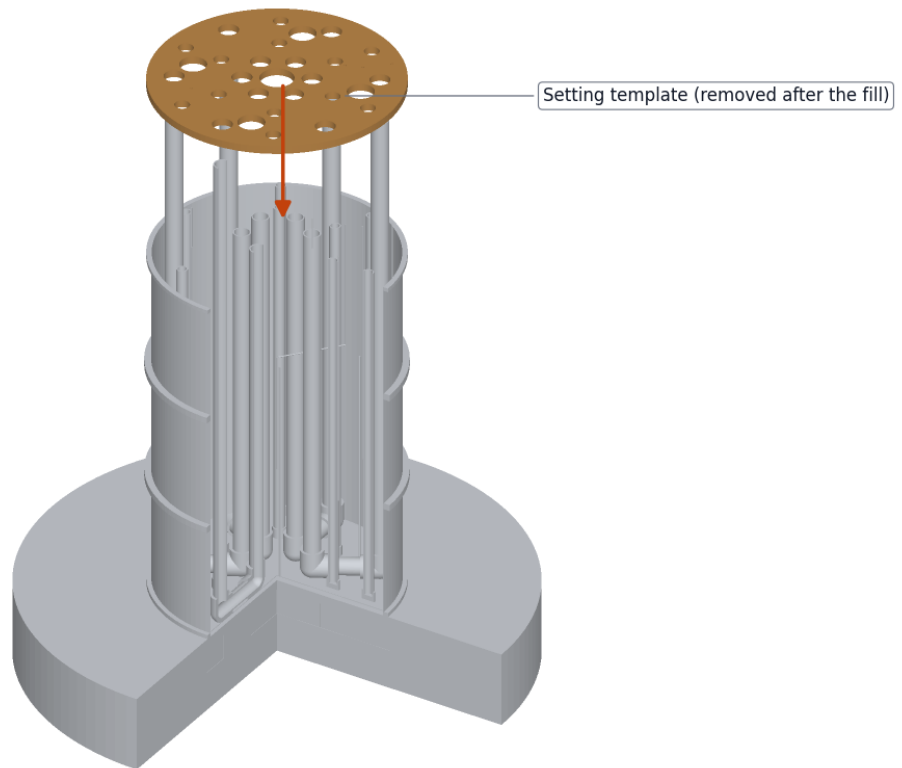
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As section 3.6: T1 to T3 wired to their wells, guide rods for T4 to T7 standing on the floor with the sheaths tied to them.

Step 7: setting template onto the rim

Step 7: setting template onto the rim

Lower it over every pipe and sheath end; clamp it to the rim at four points



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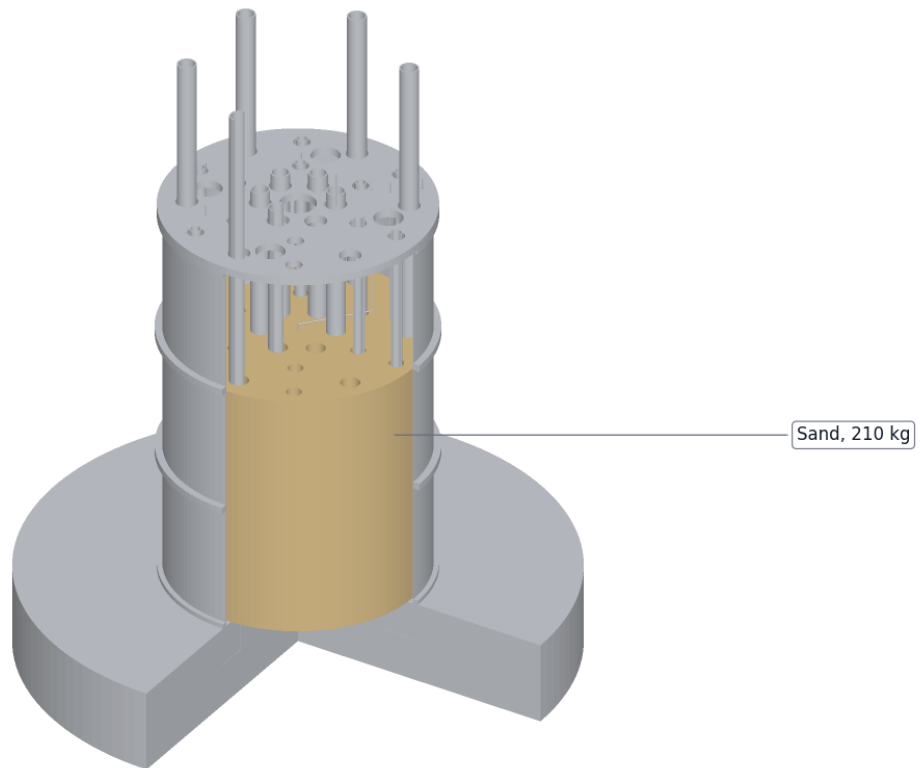
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Lower it over every pipe and sheath end, 0° marks together, and clamp it at four points. **Hold point:** every pipe stands upright in its hole; the template lies flat.

Step 8: fill with sand

Step 8: fill with sand

Pour through the template in 100 mm lifts to 571 mm, rodding each lift; weigh every bag (210 kg)



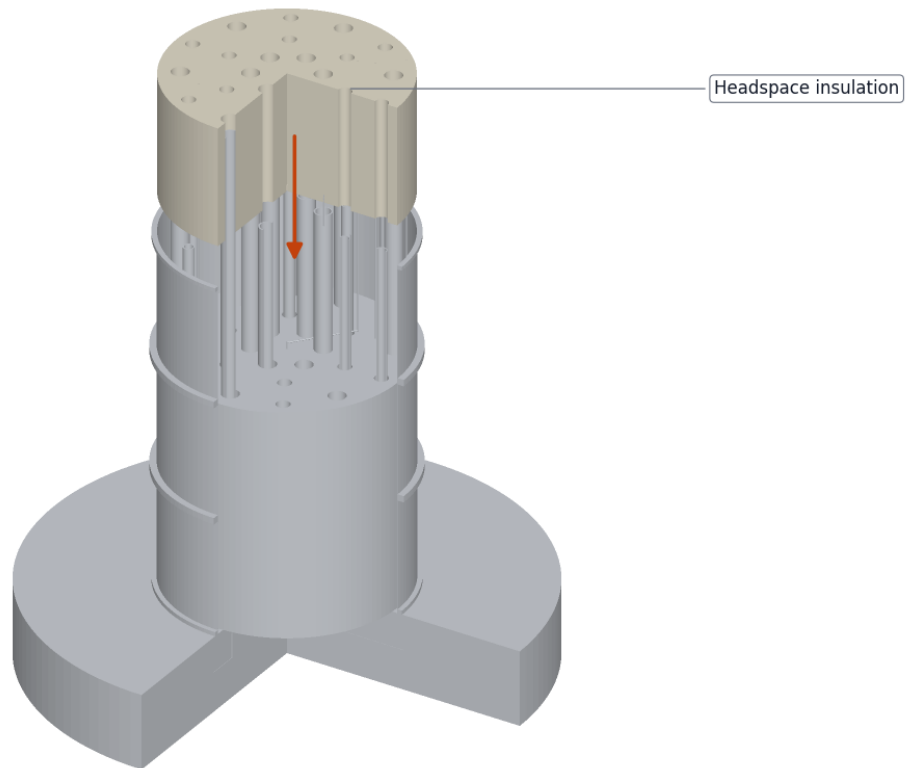
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github.com/BoujeeEnjinia1701/thermabrick

Dust control as safety stop S1: outdoors or local exhaust, and a respirator. Pour through the pour holes in 100 mm layers to 571 mm above the floor, rodding each layer round the pipes. Weigh every bag: 210 kg in all. Check the pipes stay upright as you go.

Step 9: template off, headspace insulation in**Step 9: template off, headspace insulation in**

50 mm AES blanket on the sand, then stone wool to the rim, cut round every pipe



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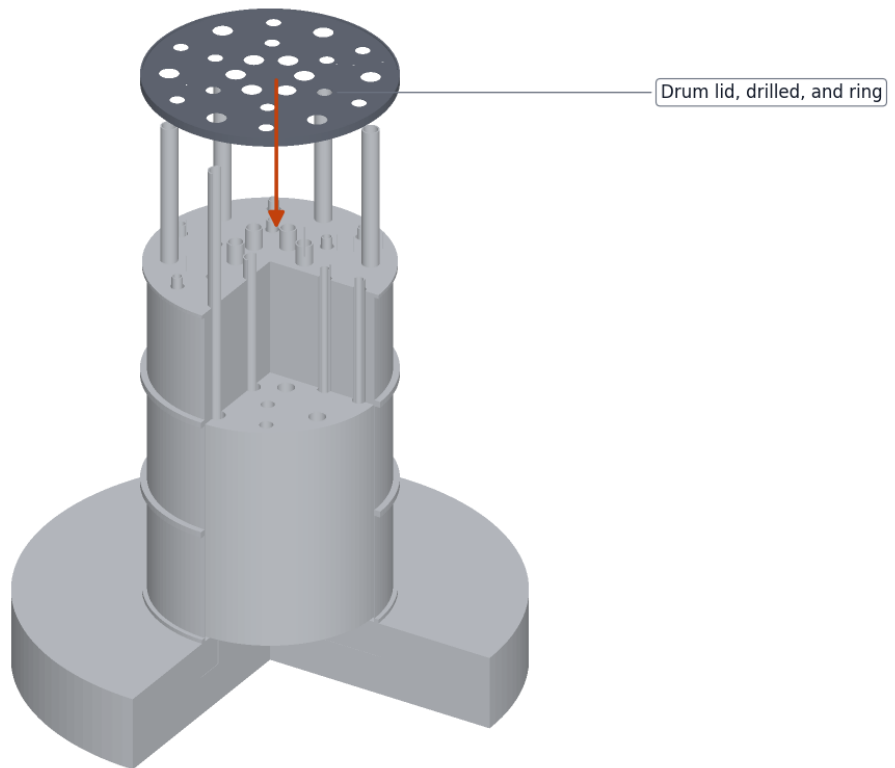
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Lift the template off. Lay 50 mm of AES blanket on the sand, then stone wool batt to the rim, cut round every pipe and sheath. Bend T4 across the sand surface first (section 3.6).

Step 10: lid and closing ring

Step 10: lid and closing ring

0° marks lined up; lower over all 24 pipes; rope collar round each; ring bolted



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

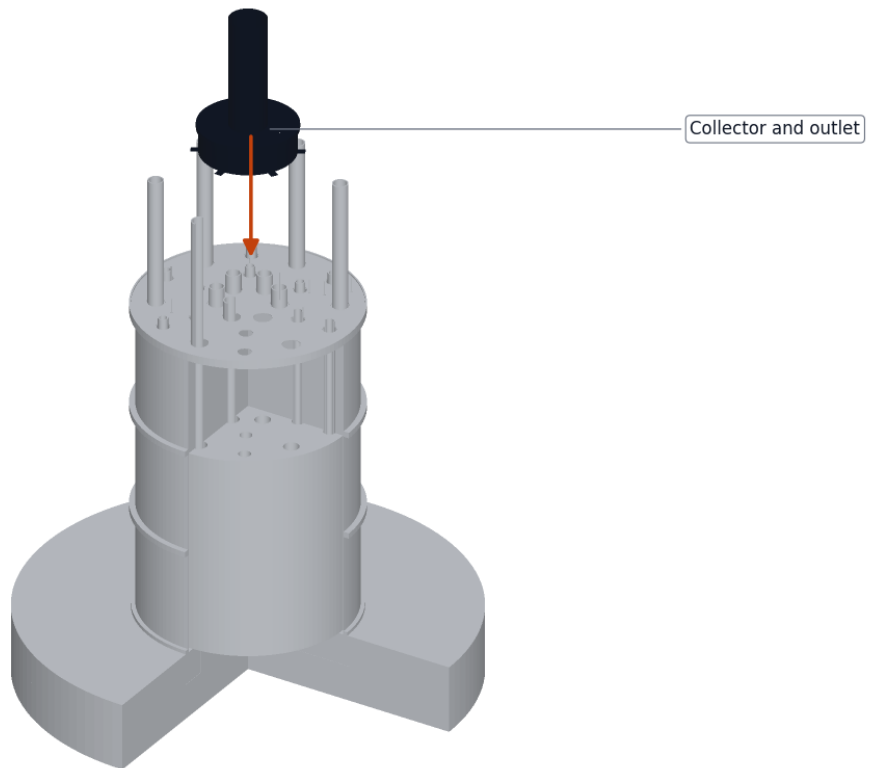
github.com/BoujeeEnjinia1701/thermabrick

0° marks lined up; lower the lid over all 24 pipe ends and the four sand sheaths, passing T1 to T3 through their well holes. Lay an AES rope collar round each pipe and bolt the closing ring.

Step 11: collector and outlet onto the lid

Step 11: collector and outlet onto the lid

Over the six short legs; tabs riveted to the lid with steel rivets; AES rope round the foot



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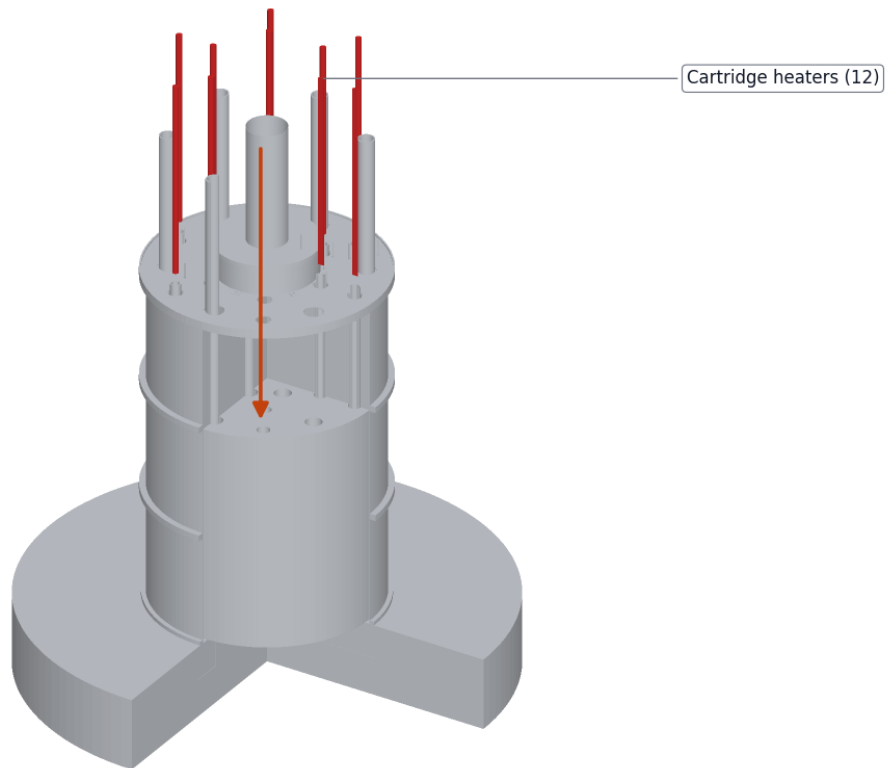
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Over the six short legs, tabs between the wells; one steel rivet through each tab into the lid; AES rope round the foot of the ring. Fit T8 through the collector top.

Step 12: heaters into the wells

Step 12: heaters into the wells

Remove the tape; lower each heater on its leads to the cap; lay the leads toward the junction box



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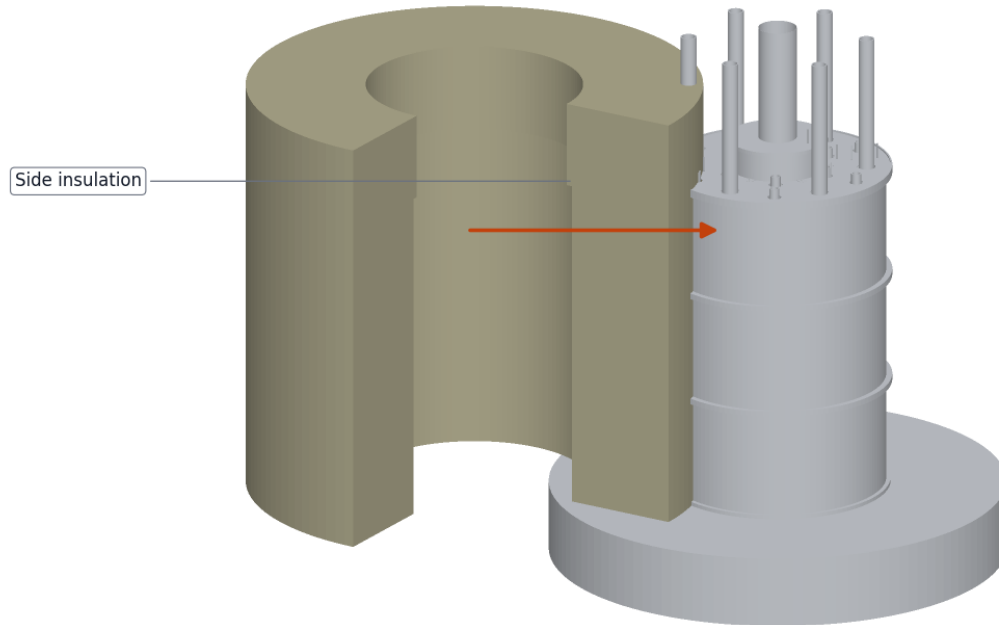
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Remove the tape. Lower each heater on its leads until it rests on the cap. Lay the leads across the lid toward the junction box position (120°), labelled by well. **Hold point:** each heater reads 1 M Ω or more at 500 V to its sheath before the top insulation covers it.

Step 13: side insulation

Step 13: side insulation

Two 25 mm AES blanket layers, then three 89 mm stone wool batt layers, joints staggered



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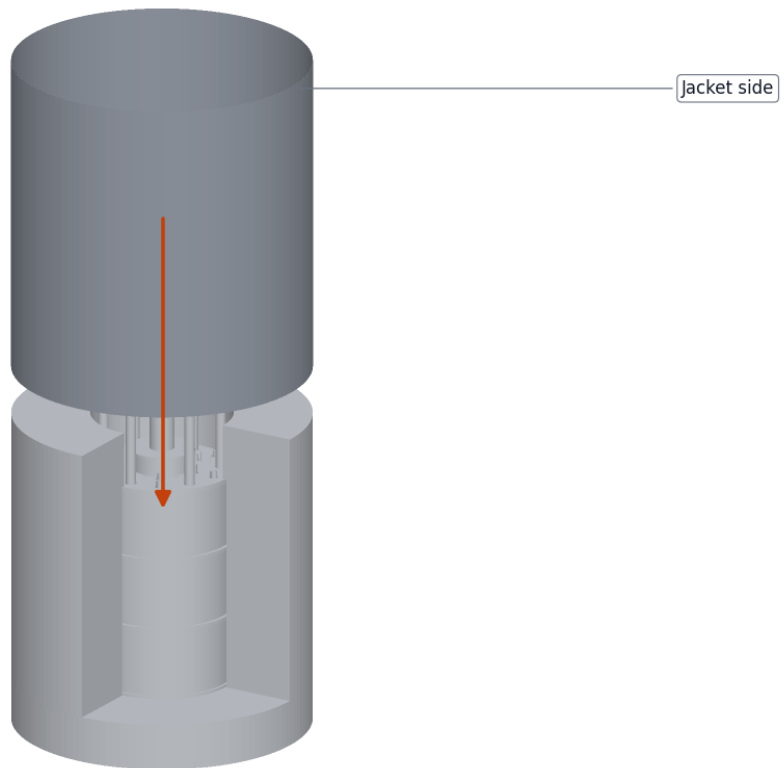
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Two layers of 25 mm AES blanket against the drum, then three layers of 89 mm stone wool batt, joints staggered, held with wire until the jacket closes.

Step 14: jacket side

Step 14: jacket side

Three rows of flashing wrapped round the insulation, strapped tight, laps riveted



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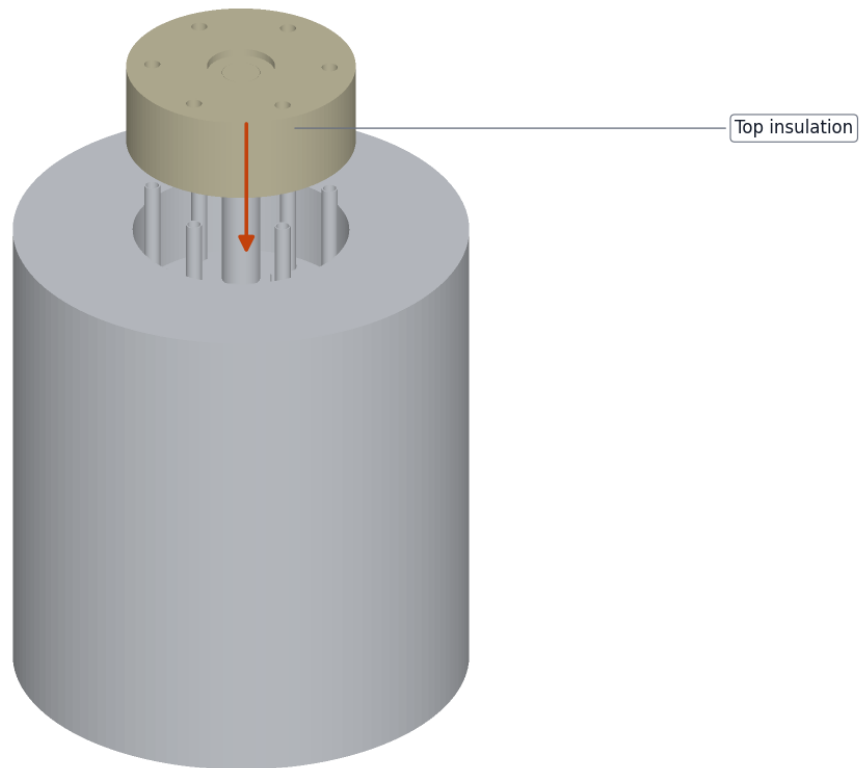
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Three rows, strapped tight and riveted as section 3.8.

Step 15: top insulation

Step 15: top insulation

50 mm AES over the lid and collector, then stone wool to the jacket top; heater leads and sheaths through it



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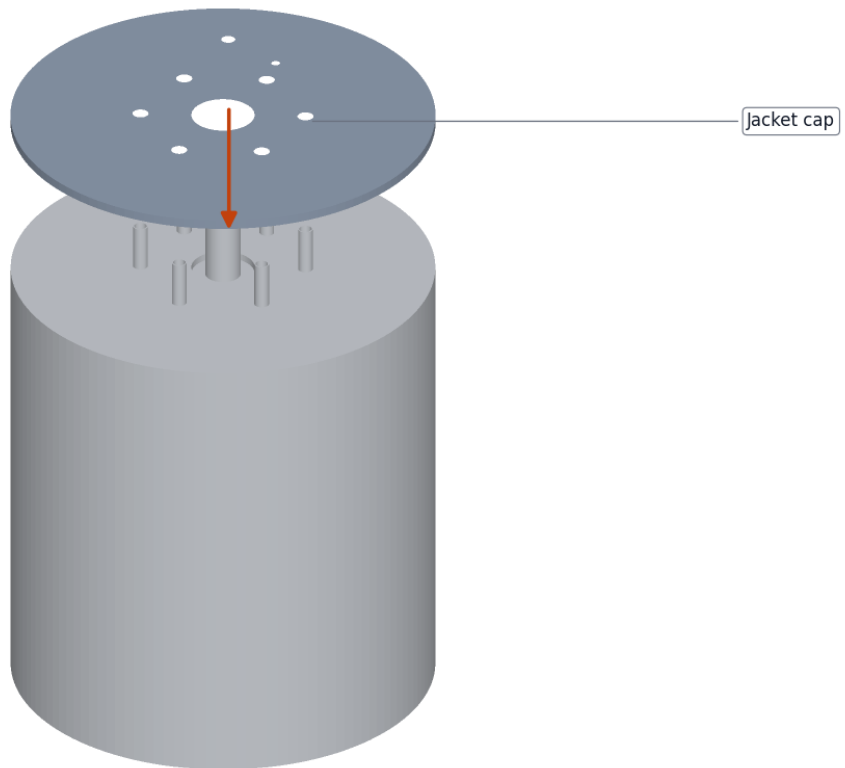
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50 mm of AES blanket over the lid and collector, then stone wool batt to the top of the jacket, cut round the outlet and the long legs. Bring the heater leads up at the junction box position and the sheaths at the exit position; coil the spare sheath in the batt.

Step 16: jacket cap

Step 16: jacket cap

Over the inlet legs and outlet; leads and sheaths through their grommets; eight screws in the skirt



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

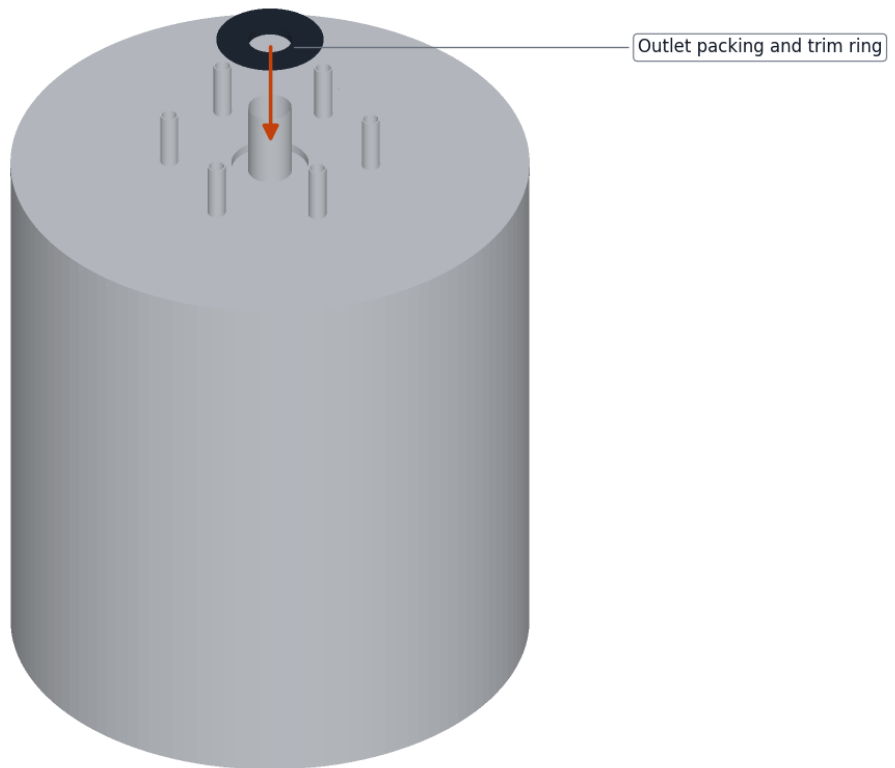
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Lower it over the long legs and the outlet, threading the leads and sheaths through their grommets. Eight screws through the skirt. Foil tape round each long leg.

Step 17: outlet packing and trim ring

Step 17: outlet packing and trim ring

Pack the 39 mm gap with AES blanket, lay the black steel ring over it, four screws



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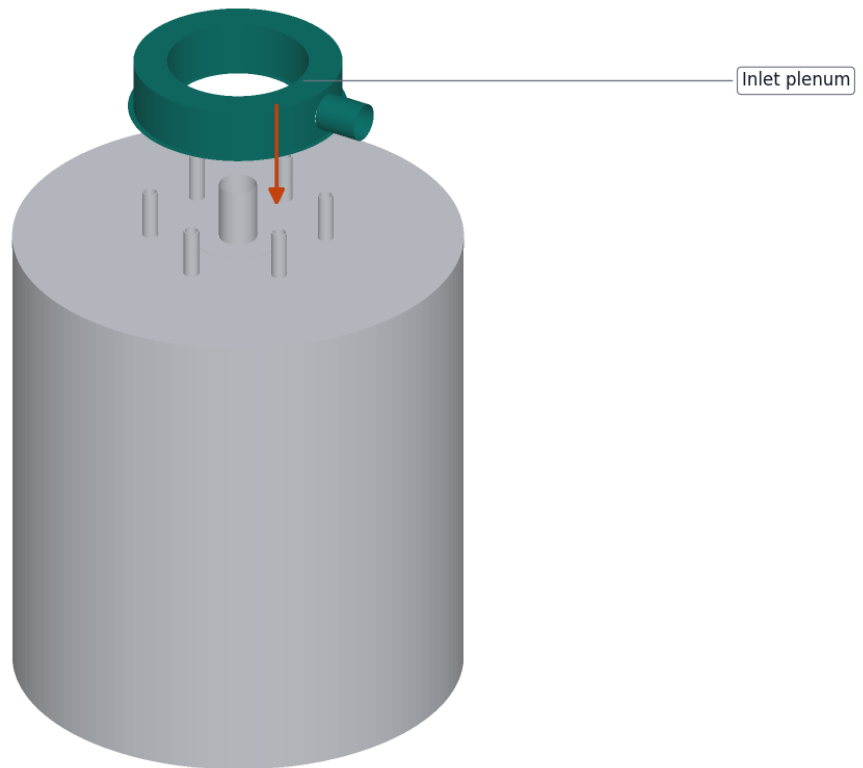
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Pack the 39 mm gap round the outlet with AES blanket, 20 mm deep. Lay the black steel ring over it and screw it to the cap.

Step 18: inlet plenum

Step 18: inlet plenum

Over the six inlet legs; flanges riveted to the cap and sealed with foil tape; collar at 0°



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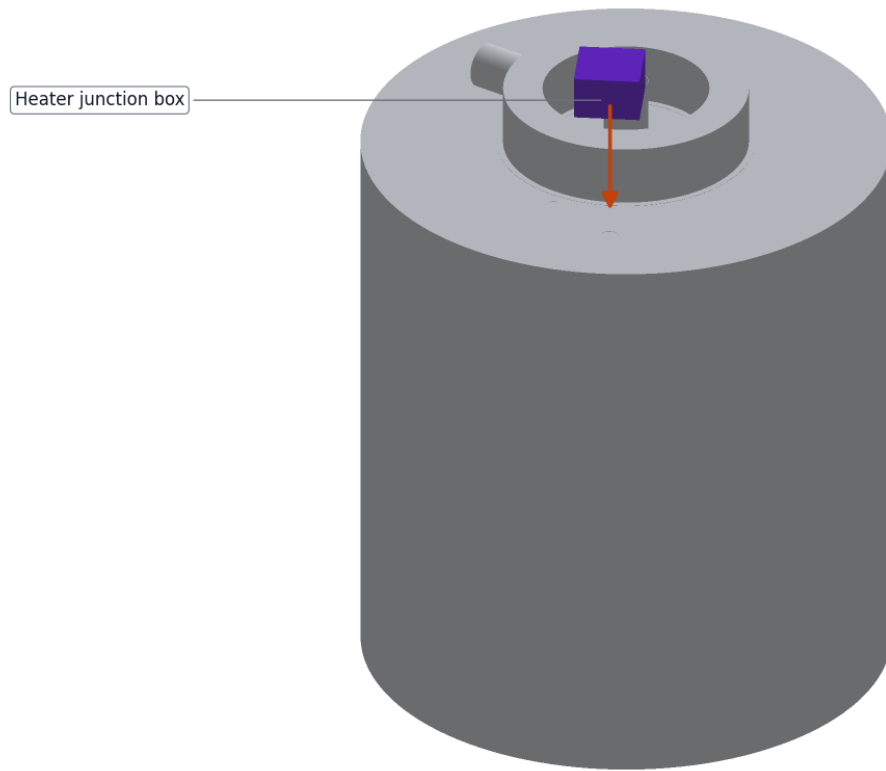
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Over the six long legs, collar at 0°; rivet the flanges to the cap and seal them with foil tape.

Step 19: heater junction box

Step 19: heater junction box

Seen from behind. Leads through the grommet into the box; land them on the ceramic terminal blocks; four screws



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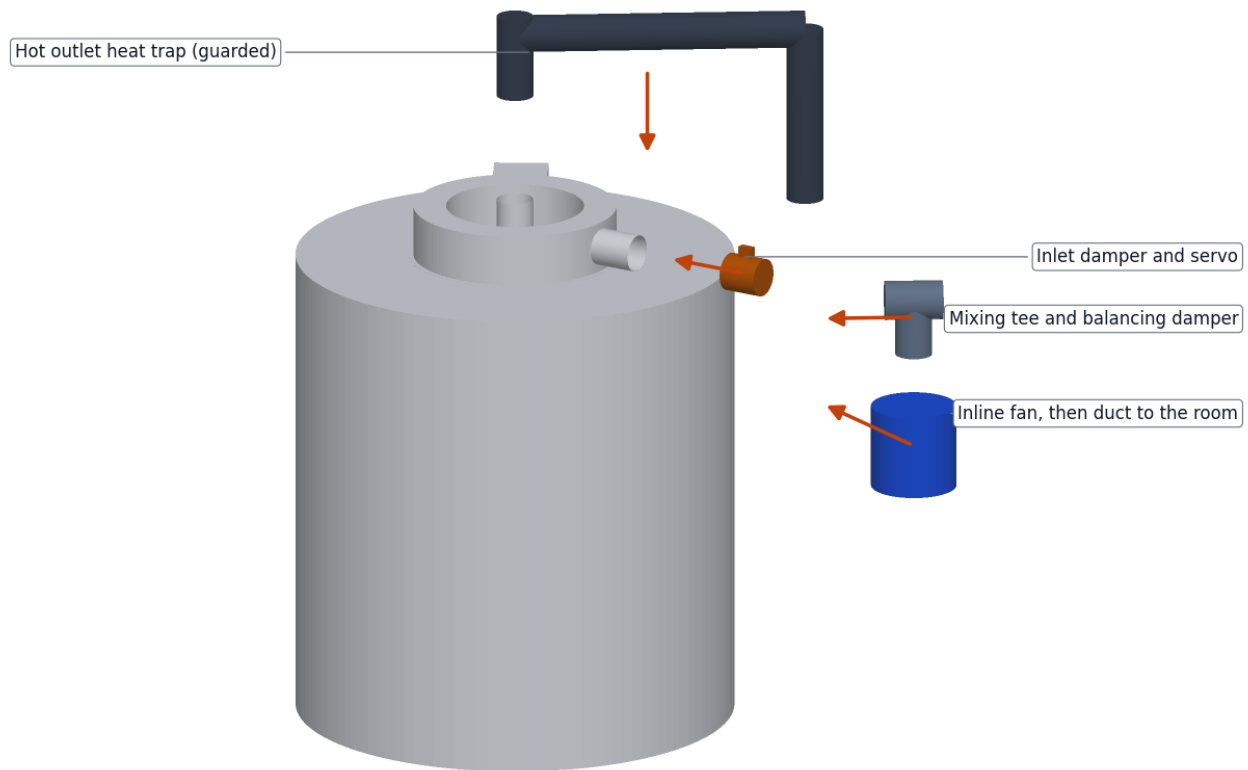
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Over the 40 mm grommet; four screws to the cap. Land the leads on the ceramic terminal blocks in two groups of six, as Figure 25.

Step 20: air path: damper, outlet duct, tee and fan

Step 20: air path: damper, outlet duct, tee and fan

Damper on the inlet collar; outlet rises and turns down to the tee; fan below the tee, then flexible duct



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The damper and its servo on the inlet collar. The 4 in outlet rises, turns and drops to the mixing tee as a heat trap; guard it with a perforated steel sleeve and keep it 450 mm from anything combustible. The balancing damper on the tee's room-air branch, then the fan below the tee, then the flexible duct to the register. The layout shown is one site; the rules are what matter.

Step 21: control enclosure and wiring

The picture for this step is Figure 25. The control enclosure goes on the wall beside the unit. A licensed electrician runs the dedicated 240 V circuit and checks the wiring of Figure 25. **Hold point:** safety stop S5.

■ 5. First checks

These are the checks a TRL 4 test report would record; this plan only lists them. Requirement numbers are those of TBK-REQ-001.

Table 2. First checks.

Check	Requirement	How	Pass when
Envelope	R14	Tape and level	1,250 mm across or less; 1,500 mm high or less with the outlet stub
Heater insulation	R11	500 V insulation tester, each heater lead to sheath, before and after the bake-out	1 MΩ or more
Branch circuit	R11	Clamp meter on each heater group at full power	Under 16 A in all; about 6.25 A per group
Independent limit	R10	Heat the T3 input (or a thermocouple simulator) past 600 °C with the heaters on	The contactor opens both legs and stays open until reset by hand
Fail-safe	R12	Reset the controller, unplug a thermocouple, cut the export signal	Heaters off and damper closed within 60 s each time
Charge follows export	R2	Log export and heater power through a sunny day	Heater power tracks surplus in 25 W steps; little or no grid import
Well wall and sand limit	R8	Log T1, T2 and T4 to T7 through a full charge; a temporary thermocouple on the drum wall beside an outer well	Well walls 550 °C or less; drum wall 500 °C or less
Jacket and cap surfaces	R9, R13	Contact probe at full charge: jacket side and top, and the cap at the edge of the outlet hole	Jacket 45 °C or less; cap edge under 200 °C
Supply air	R6	Log the supply air sensor through a discharge	55 °C or less at all times
Stored heat	R1	Calorimetric discharge from full (air flow and temperature rise)	18 kWh(th) or more between 150 and 450 °C
Heater service	R17	Lift the cap and the top insulation, draw one heater, refit it	Done without disturbing the sand

■ 6. Safety stops

Stop at each point. Carry on only when everything listed is true.

- **S1. Before cutting firebrick or pouring sand.** Work outdoors or with local exhaust; P100 or N95 respirator, eye protection and gloves on; the area wet-cleaned afterwards, never dry-swept.
- **S2. Before handling fiber insulation.** Gloves, long sleeves, eye protection and a respirator; offcuts bagged.
- **S3. Before burning out the drum.** Outdoors, clear of buildings, with water to hand; the drum empty, open and never sealed.
- **S4. Before the top insulation covers the heaters.** Every heater reads 1 MΩ or more at 500 V lead to sheath; every lead is labelled and its beads are whole.
- **S5. Before the 240 V circuit is energized.** A licensed electrician has run and checked the dedicated circuit; the enclosure is closed and earthed; the junction box lid is on; the high limit trips the contactor in a test with the heaters disconnected.

- **S6. Before the first firing (bake-out).** The room is ventilated and smoke detectors are not disabled; the hot outlet guard is fitted and nothing combustible is within 450 mm of it; someone stays with the unit until it holds 150 °C steadily.
- **S7. Before any heater is pulled or the lid opened.** The sand mean reads below 60 °C; the circuit is isolated and locked off.

■ 7. Tools, skills and workspace

Tools. Pipe threader for 3/4 and 1-1/4 in (rental) with cutting oil; pipe cutter or hacksaw; two pipe wrenches; reamer; bi-metal hole saws 6 to 106 mm and a drill with a side handle; step drill; aviation snips; hand seamer (folding pliers); rivet gun for 4.8 mm rivets; sheet metal screwdriver bits; coarse handsaw or masonry blade for firebrick; long serrated insulation knife; trammel or string and nail; tape, steel rule and square; two ratchet straps; flap disc and angle grinder for paint stripping; 500 V insulation tester; multimeter with a type K input; clamp meter; kitchen or platform scale for the sand bags.

Skills. Pipe cutting and threading, sheet metal layout, cutting, folding and riveting, careful handling of fiber and dust. The 240 V circuit and enclosure wiring need a licensed electrician; everything else needs no certified trade.

Workspace. The final position on a concrete slab inside the heated space, with about 1 m of clear floor round it; an outdoor area for burning out the drum, cutting brick and pouring dusty materials; a bench with a pipe vice.

Personal protective equipment. P100 or N95 respirator for sand, brick and fiber; safety glasses; cut-resistant gloves for sheet metal; leather gloves for the drum burn-out; hearing protection for grinding.

■ 8. Where the numbers come from

- Model and constructability checks: `cad/src/model.py` (`python cad/src/model.py --check` , 72 checks); STEP and STL in `cad/step/` and `cad/stl/` .
- Pictures: `cad/src/build_plan_media.py` , using `.kit/build_views.py` ; written to `docs/05-build-plan/` and `cad/drawings/TBK-DWG-101` to `TBK-DWG-110` .
- General arrangement: `cad/drawings/TBK-DWG-001.pdf` , Rev P2.
- Calculations: `docs/04-calcs/TBK-CAL-001-sizing.md` (TBK-CAL-001 v0.3) and `docs/04-calcs/tbk_cal_001.py` ; mass and floor load section 7, electrical section 6.
- Bill of materials: `bom/bom.csv` and `bom/bom-notes.md` .
- Decisions: `docs/decisions/0003-design-for-construction.md` (TBK-DDR-003), with TBK-DDR-002; open items in `docs/06-design-decisions.md` (TBK-DEC-001).
- Requirements: `docs/03-requirements.md` (TBK-REQ-001 v0.10).